University of Engineering and Technology, Lahore
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Department of Computer Science & Engineering

VidCraft

Intelligent Prompt-Driven Media Generation Pipeline

Software Requirements Specification

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Plagiarism Certificate

Project: VidCraft: Intelligent Prompt-Driven Media Generation Pipeline

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Abstract

VidCraft takes a user's written idea through a creative brief, a storyboard and a viewable image or short video. Prompt analysis, clarification questions, cinematography retrieval and cooperating software agents address different stages. The analysis considers subject clarity, action specificity, environmental detail, visual richness and temporal coherence. These five measures help identify what needs clarification before shot planning and media generation.

The project contribution includes scoring rules for visual prompts, contradiction detection, a clarification strategy, shared world state, agent coordination and a curated cinematography knowledge base. Quality feedback is limited to a defined number of attempts. Programmatic rendering turns structured scene descriptions into compositions, and media adapters connect suitable generative models. Selected shots, narration, captions and post-processing share a timeline. Background processing and recovery at individual stages help the system work within limited computing resources.

VidCraft connects pretrained generative models through interchangeable hosted and local adapters. Hosted inference is the primary mode; Local Demo supports experiments with smaller models and, if developed and evaluated, a task-specific fine-tuned adapter. Reproducible experiments are planned to assess prompt quality, preservation of intent, storyboard coherence, rendering correctness and resource trade-offs. This SRS sets out the behaviour, interfaces, data, quality requirements and acceptance tests against which the project will be assessed.

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines VidCraft's required behaviour for the team, testers and examiners. It covers user actions, component interactions, stored information and the conditions that the completed project must meet. The requirements describe the intended behaviour of the system. They do not imply that every feature has been implemented or evaluated.

1.2 Objectives

1. Assess the detail in an input prompt and explain where the user can improve it.
2. Resolve ambiguity without changing the subject, action or setting explicitly chosen by the user.
3. Plan a coherent shot sequence using specialized agents and techniques retrieved from the cinematography knowledge base.
4. Carry shared scene details and the chosen style through planning and rendering.
5. Produce images and short videos using suitable generative media components together with programmatic compositions.
6. Manage background jobs, narration, captions and delivery with limited retries and understandable progress updates.
7. Support hosted and local inference within available resources and measure the pipeline's contribution.

1.3 Problem Statement

A visual idea may leave the subject, action, surroundings, framing or timing unclear. Generating media directly from that idea can give a generic result or leave out details the user considers important. When a narrative contains several events, someone must also decide their shot order, camera treatment and shared scene details. Repeated generation offers little control over these decisions and can consume additional time and resources.

VidCraft separates this work into analysis, clarification, planning, generation and verification. Its components establish a creative plan and keep the outputs of individual models connected to that plan. The project examines whether this approach gives users clearer feedback and better control over prompt-driven media creation while remaining practical within the computing budget available to the student team.

1.4 Project Contribution

The team's work concerns the analysis, planning and coordination that carry a user's idea through to a media artifact. The main contributions are:

- **Linguistic analysis:** five scoring dimensions, missing-detail detection and a visual-prompt contradiction dictionary.
- **Interactive refinement:** a limited clarification round and a structured brief that retain the user's explicit constraints.
- **Agent coordination:** defined Screenwriter and Cinematographer roles, validation between stages, semantic checks and limits on plan revision.
- **Cinematography retrieval:** a selected technique corpus, its embedding/index pipeline and retrieval of guidance relevant to each shot.
- **Continuity representation:** shared characters, setting and style alongside each shot's action and camera instructions.
- **Rendering and orchestration:** procedural compositions, job management, provider adapters, failure isolation and media assembly.
- **Synchronized delivery:** one timeline aligning retained shots, measured narration beats and captions.
- **Experimental evaluation:** controlled comparisons of refinement, agent planning and hosted/local inference, with separate documentation of any fine-tuning experiment undertaken.

Pretrained foundation models supply language and visual synthesis capabilities. The team designs, implements and evaluates VidCraft components that organize and apply those capabilities. Training a foundation model from scratch falls outside the project scope.

1.5 Intended Audience and Reading Suggestions

Supervisors and examiners should start with the objectives, scope, requirements and acceptance criteria. Developers should consult interfaces, data models, use cases and analysis models during implementation. Testers can organize evidence through requirement IDs and the test matrix. Product functions and use cases explain the workflow to users. Environment, configuration, privacy and recovery requirements are particularly relevant to operators.

1.6 Product Scope

The scope covers text input, optional dictation, prompt analysis, clarification, style settings, single-image generation, multi-shot planning and rendering. Users can follow progress, preview or download results and return to prompt history. Shared scene records, cinematography retrieval, intent checks, limited visual feedback and optional synchronized narration support these functions. Post-processing covers captions, poster frames, compression and selected colour treatment.

The deliverable is a browser-based academic prototype for controlled local or laboratory use. It excludes public account registration, subscriptions, collaborative editing, full professional video editing, native mobile applications and social-media publishing. Programmatic compositions produce stylized animation, while generated stills may be assembled with camera motion. Genuine generative video requires an enabled route with a video-capable model. Evaluation distinguishes these output types.

1.7 Document Conventions

The word “shall” states a requirement, “should” states a recommendation, and “may” indicates an optional capability. FR, NFR, OR, UC and TC refer to functional requirements, nonfunctional requirements, other requirements, use cases and test cases. High priority covers essential workflow and integrity requirements; Medium covers supporting functions; Optional applies only when the capability is enabled. Requirement and test IDs are unique within version 2.0. Figures and tables are numbered automatically.

ISO/IEC/IEEE 29148:2018 informs the requirements structure used here [1]. It serves as a reference for requirements engineering; the document does not claim certification of conformity with that standard.

Data flow diagrams use Gane–Sarson notation [10]: rounded rectangles are processes, shaded rectangles are external entities, open-ended rectangles are data stores and labelled arrows are data flows. Use-case, sequence, activity, state-transition, component and deployment diagrams follow UML 2.5 [9]. The entity–relationship diagram uses crow’s-foot notation. All diagrams use the same scale and lettering size. Their connectors are straight lines, drawn horizontally or vertically wherever the notation allows, and no connector crosses another.

2 Overall Description

2.1 Product Perspective

VidCraft divides media authoring among several components. The browser provides the studio. The backend validates requests, stores records and schedules jobs. Linguistic analysis and agent planning run in the AI service, while the rendering layer creates and processes media. Persistent application records, retrieval knowledge and generated files are stored by role.

Each stage passes structured information to the next: analysis records, creative briefs, world state, shots or narration beats. These records allow the system to check outputs, limit retries and compare alternative executions. They also retain successful work if a later stage fails.

2.1.1 Context Model

The Creator submits an idea and style choices, then receives analysis, progress updates and media. The Operator prepares local resources and configures components. The Evaluator reviews repeatable experiments and their results. Model capabilities come from hosted services or a local inference runtime, each accessed through a controlled interface.

Figure 1 is the context diagram (Level 0 data flow diagram). The whole system appears as the single process 0, surrounded by its four external entities: E1 Creator, E2 Operator, E3 Evaluator and E4 Model Services. E4 represents the hosted model services in Standard mode and the prepared local inference runtime in Local Demo mode; the selected execution profile decides which one serves a task. Each arrow is one named data flow, and no data store appears at this level.

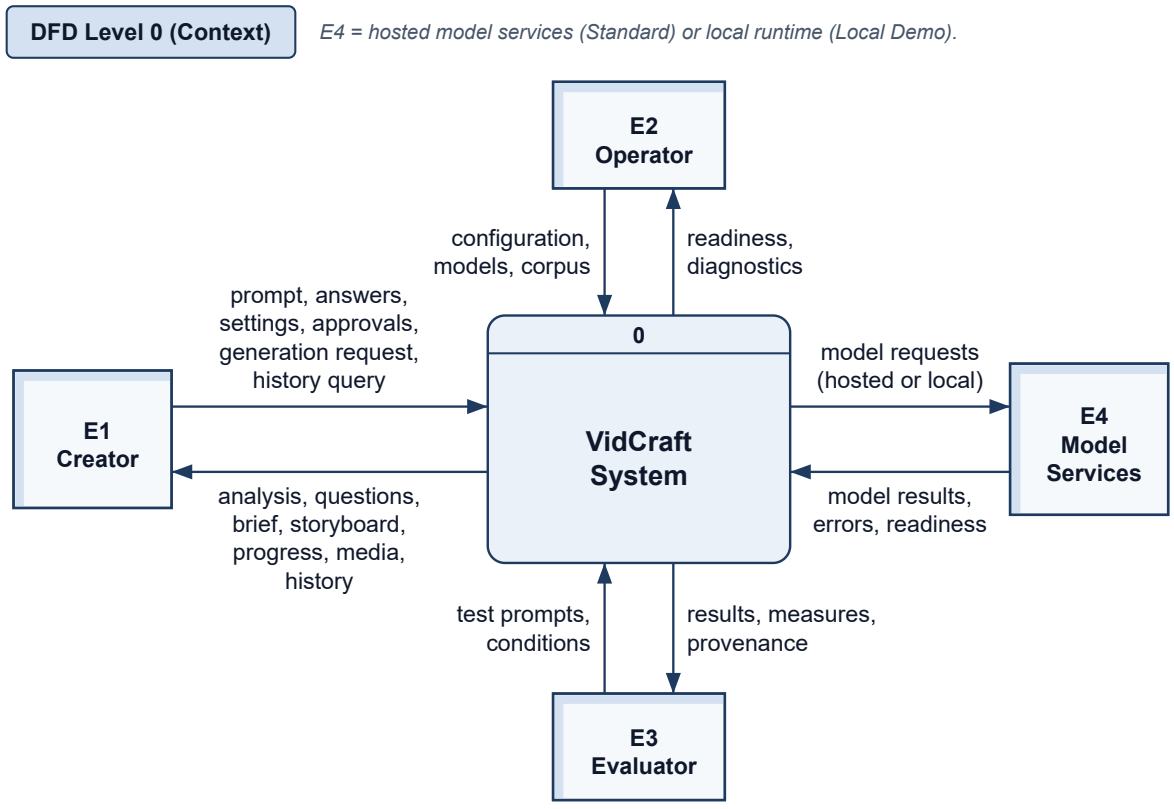


Figure 1: Context diagram (DFD Level 0) of VidCraft

2.2 Product Functions

1. **Prompt analysis:** explain clear elements, missing details and contradictions in visual descriptions.
2. **Prompt refinement:** collect focused answers and form a structured brief for review.
3. **Style control:** carry the chosen style, mood, lighting, palette and supported dimensions forward.
4. **Narrative planning:** order storyboard shots and refine camera instructions using retrieved cinematography guidance.
5. **Media creation:** produce procedural compositions or use the selected compatible route to generate media.
6. **Quality control:** check intent and shared scene details, with limited visual critique when enabled.
7. **Delivery:** combine usable shots, align optional speech/captions and provide downloadable results.
8. **Review and experimentation:** preserve prompt versions and compare standard and local runs under recorded conditions.

2.2.1 Data Flow Diagrams

Validation, scoring and optional clarification turn the input prompt into a creative brief. The chosen style and execution settings travel with it. A single-image request goes to a compatible image route. A storyboard request first passes through narrative planning, technique retrieval and intent checking, then enters the generation queue. Usable shots proceed to narration, composition and post-processing. Results and metadata remain linked to their source prompt and generation attempt.

The Level 1 diagram (Figure 2) decomposes process 0 of the context diagram into six interlinked processes. Processes 1.0 to 4.0 form the main pipeline and pass their results through the data stores placed between them. Processes 5.0 and 6.0 connect to the same stores. The diagram is balanced with Level 0: it has the same four external entities, and each Level 0 flow appears in it, either unchanged or divided into its component flows (for example, *media* becomes *image* from 3.0 and *video, downloads* from 4.0). Where a store or entity is needed in several rows, it is drawn again with the Gane–Sarson duplicate mark so that no flow crosses another. Table 1 defines each process and data store.

Table 1: Processes and data stores of the Level 1 data flow diagram

ID	Name	Responsibility or content
1.0	Analyze and Refine Prompt	Validates and stores the prompt, scores it, asks clarification questions and builds the confirmed brief (FR-01 to FR-06).
2.0	Plan Storyboard	Applies the selected style and workflow, drafts and validates the shot plan, retrieves techniques, checks intent and presents the storyboard (FR-07, FR-09, FR-11 to FR-15).

Continued on next page

Processes and data stores of the Level 1 data flow diagram (continued)

ID	**Name**	**Responsibility or content**
3.0	Generate Media	Admits and queues the job, validates route and dimensions, generates and checks images or shots, and records attempts (FR-08, FR-10, FR-16 to FR-23, FR-29).
4.0	Assemble and Deliver Media	Adds optional narration, assembles the common timeline, post-processes and delivers the artifacts (FR-24 to FR-28).
5.0	Manage History and Evaluation	Answers history and comparison queries and runs controlled evaluations on frozen test prompts (FR-30, FR-31, FR-40).
6.0	Configure and Monitor System	Applies the operator's configuration, prepares the index and reports model and service readiness (FR-32 to FR-39).
D1	Prompt Records	Prompts, prompt versions, analyses and confirmed briefs.
D2	Plans and Jobs	Storyboards, world state, shots, generation attempts and job states.
D3	Technique Index	Cinematography corpus, FAISS index and encoder metadata.
D4	Media Files	Shot assets, the master video and its derivatives.
D5	Configuration	Server-side routes, limits, feature flags and execution profiles.

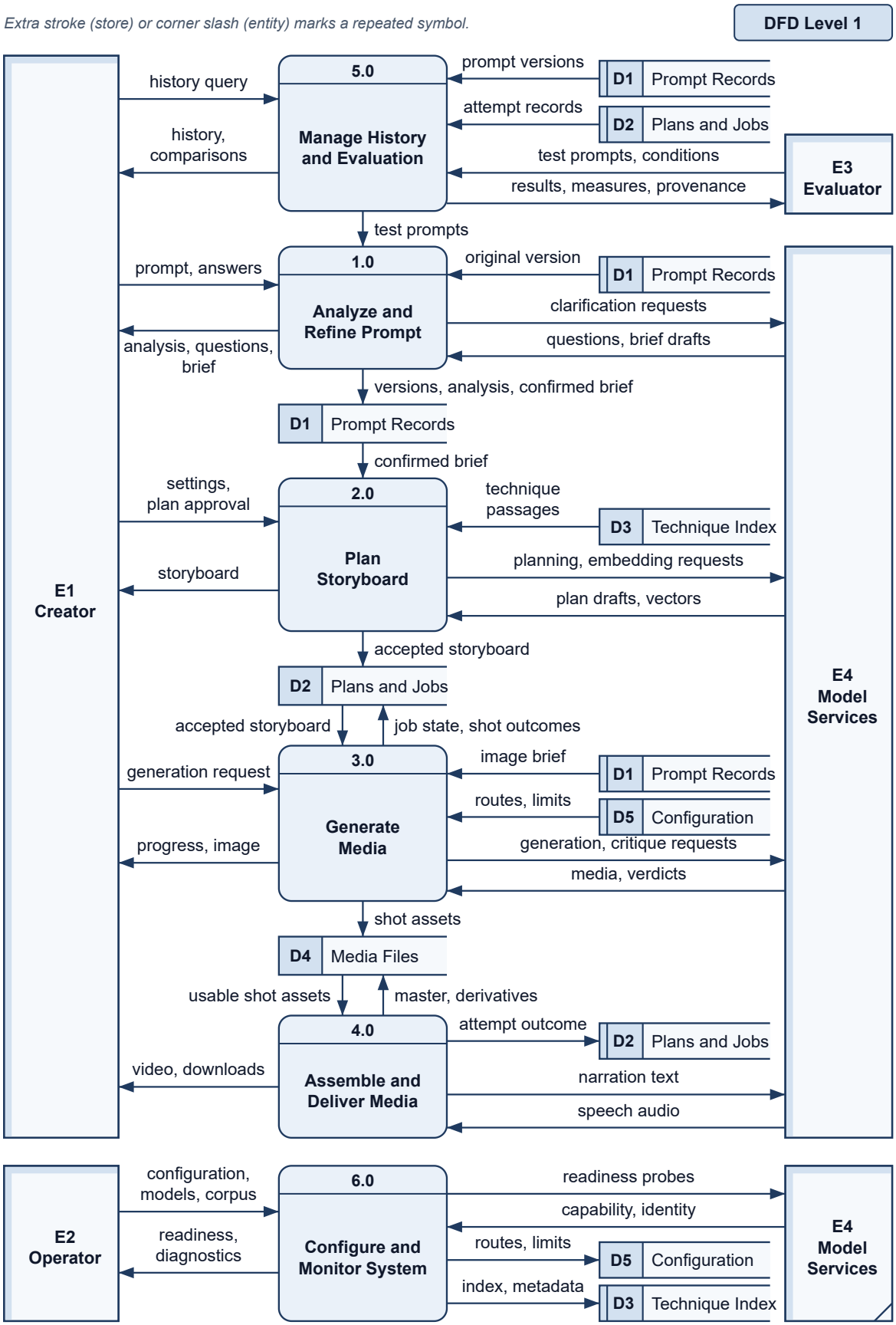


Figure 2: Data flow diagram, Level 1

The Level 2 diagrams decompose the two processes that contain the team's main analysis and planning work. Each balances with its parent: every flow between a sub-process and an entity, store or other process corresponds to a flow of the parent process in Figure 2, or to one of its component flows.

In Figure 3, process 1.1 validates and stores the original prompt, and 1.2 calculates the five dimension scores and flags. Process 1.3 generates at most six clarification questions for unresolved details. Process 1.4 combines the answers with the stored original version to form the confirmed brief. Test prompts from process 5.0 enter at 1.1 in the same way as a Creator's prompt.

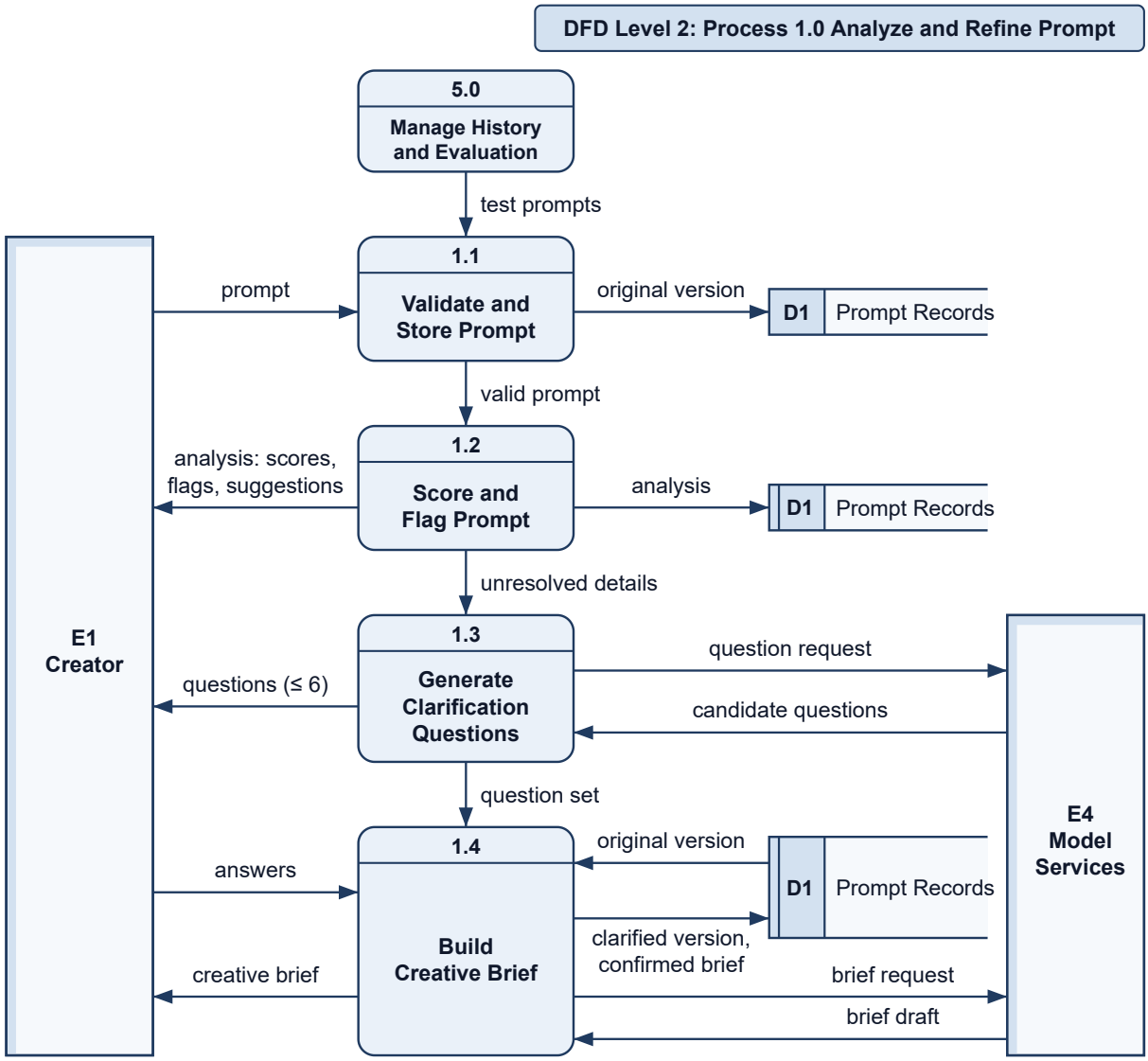


Figure 3: Data flow diagram, Level 2: process 1.0 Analyze and Refine Prompt

In Figure 4, the Screenwriter (2.1) drafts three to five shots and the world state, and 2.2 validates their structure. The Cinematographer (2.3) retrieves technique passages and refines camera instructions. Process 2.4 compares the refined narrative with the confirmed brief, and 2.5 presents the storyboard for acceptance. Both feedback flows into 2.1 are bounded by the same limit of a first draft and at most two revisions (FR-13); a plan that remains invalid is blocked from rendering (FR-11).

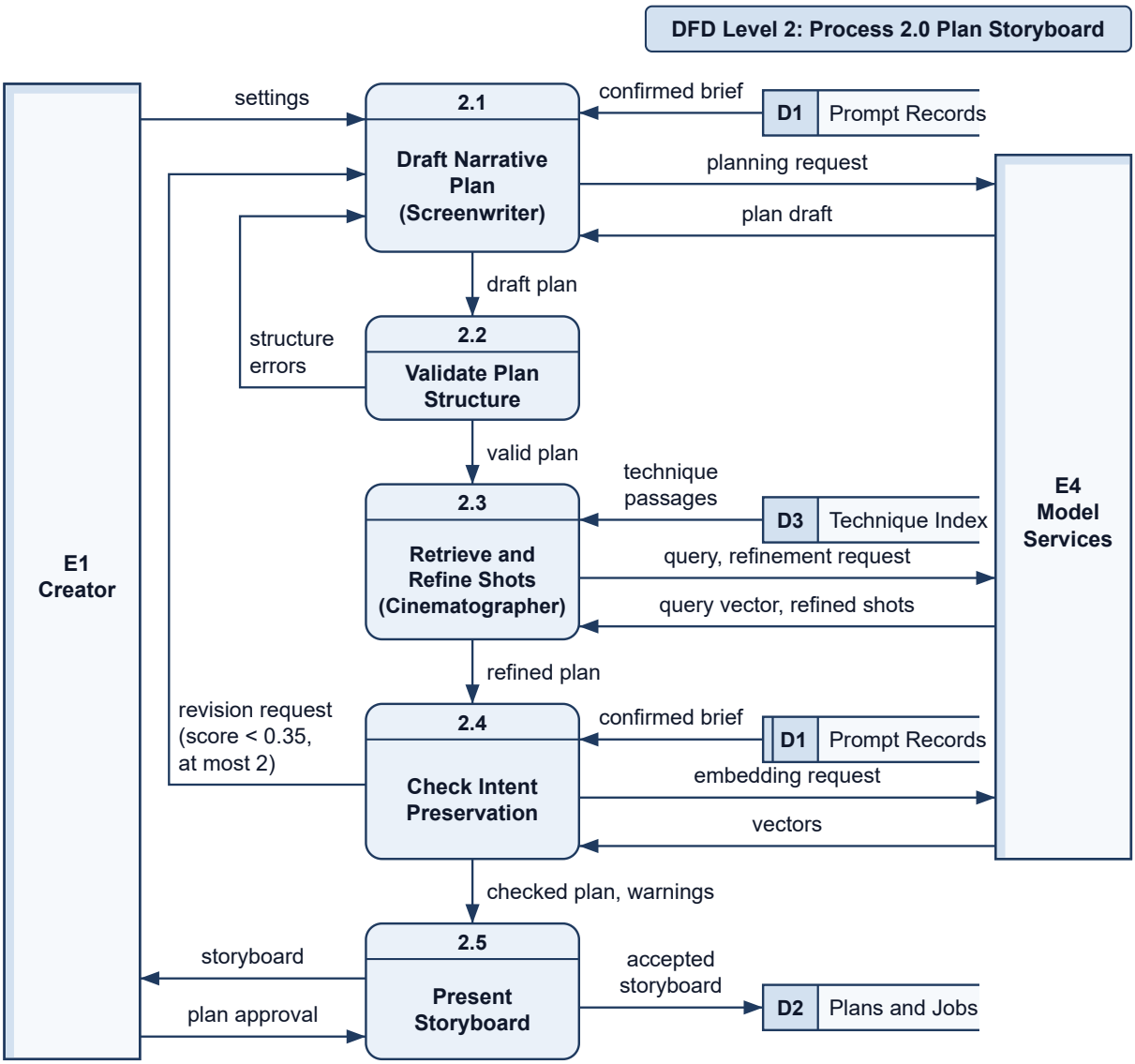


Figure 4: Data flow diagram, Level 2: process 2.0 Plan Storyboard

2.3 User Classes and Characteristics

Table 2: User classes and characteristics

User class	Characteristics	Main responsibilities
Creator	Student, educator or content creator familiar with browsers.	Submit ideas, answer questions, choose settings, review the plan and download results.
Operator	Team member familiar with deployment and configuration.	Set up services, models and indexes; maintain credentials, resource limits and retained data.
Evaluator	Supervisor, tester or researcher examining system behaviour.	Check requirements, conduct controlled comparisons and review evidence alongside its limitations.

2.4 Operating Environment

The client requires a desktop or laptop browser supporting the application's image/video formats. Its layout shall also accommodate mobile-width screens. A microphone is optional. The planned stack comprises React/TypeScript, Node.js/Express, Python/FastAPI [8], MongoDB, Redis/Bull, spaCy [2], LangGraph [3], FAISS [5], Remotion [6] and FFmpeg [7]. Section 3 describes their interfaces.

The acceptance-test reference setup uses a four-core development machine, 16 GB RAM and at least 20 GB free disk space. This is a test configuration, not a proven minimum for every local model. Performance reports must record the actual CPU, RAM, operating system and any GPU used. The primary hosted profile requires no dedicated GPU. Suitable local models must be chosen according to measured capacity, model size and quantization.

Development services may run on Windows with WSL or a compatible Linux environment. Enabled hosted services and initial resource downloads require network access. After preparation, the local profile may run without hosted inference if every enabled processing component is local. The baseline language support covers stored English prompts and processing through the English NLP pipeline.

2.5 Design and Implementation Constraints

- The available computing and generation budgets require limits on concurrent jobs, retries and model calls.
- Agent output must pass its structured schema checks before it can be stored or rendered.
- The retrieval index and its query embeddings must share the same model/vector space.
- Style and scene constraints explicitly set by the user must pass intact between components.
- Credentials and other secrets must stay in server configuration, outside browser code.
- The application controls shared scene parameters; visual continuity in generated media remains best-effort.
- Local experiments use existing small models or a documented adaptation; they exclude foundation-model training.

2.6 Assumptions and Dependencies

The operator supplies accessible model resources, required credentials, sufficient storage and a working database/queue. VidCraft cannot control changes in model availability, network access or external quotas. It shall retain user work and explain failures caused by those dependencies. Users supply text they are permitted to process. Any training or evaluation dataset must have documented provenance.

2.7 System Architecture and Agent Responsibilities

The Prompt Analyzer applies deterministic linguistic scoring. Clarification asks a limited set of questions and produces the brief. The Screenwriter builds the narrative plan; the Cinematographer retrieves techniques and refines camera/style instructions. A semantic gate checks intent preservation. When enabled, the Critic reviews eligible generated visuals and may request only limited revisions. Narration and rendering turn the accepted plan into an audiovisual result with a defined timeline.

The backend handles provider selection, queue submission and stored progress. Agents cannot silently switch to a more expensive provider. Rendering uses the shared scene context and one rule for including shots. Every result remains traceable to the inputs, settings and model execution profile used to create it.

Figure 5 shows these components as a UML component diagram arranged in tiers: the browser client, the Node.js gateway and generation worker, the AI service and the rendering pipeline, and finally the data stores and model providers. Dashed arrows are usage dependencies labelled with their purpose or protocol. All model calls from the AI service pass through the inference adapters, which select the hosted or local route recorded in the task's execution profile.

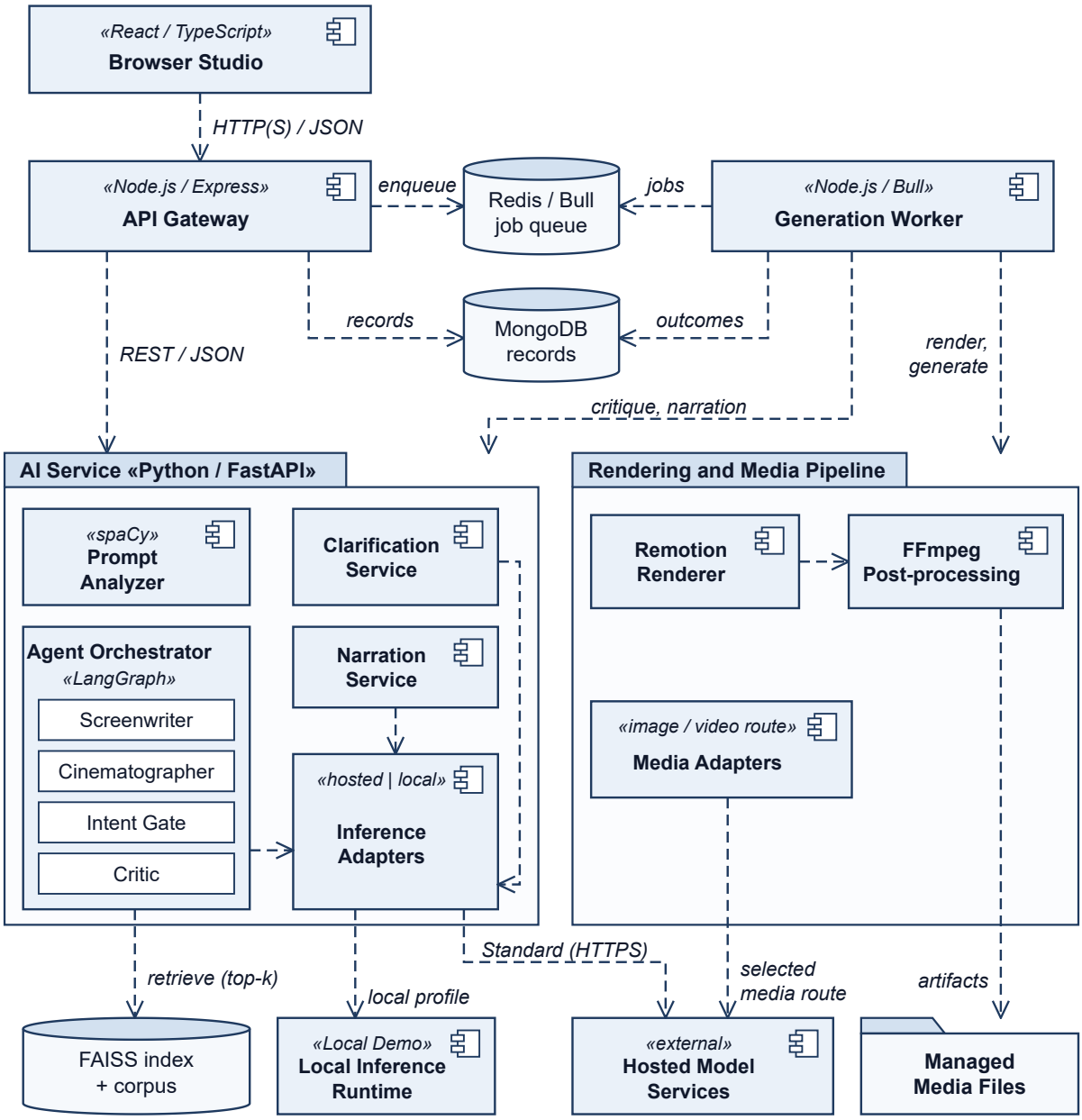


Figure 5: Component architecture of VidCraft (UML component diagram)

3 External Interface Requirements

3.1 User Interfaces

The interface shall provide these workflow views or panels:

- **Prompt composer:** text entry, examples or guidance, Analyze and optional microphone control.
- **Analysis and refinement:** the overall score, five dimensions, flags, suggestions, clarification questions and a brief for review.
- **Style and generation settings:** choices for style, mood, lighting, palette, custom terms, supported aspect ratio, workflow and media route.
- **Execution control:** a compact *Standard / Local Demo* switch explaining the chosen inference mode and whether it is ready.
- **Storyboard review:** shot order, descriptions, camera instructions, durations and shared details before generation.
- **Progress and results:** the active stage, shot status, partial errors, image/video previews and available downloads.
- **History:** stored prompts, their revisions and generation attempts, with access to comparisons and results.

The selected mode shall stay visible. Standard normally uses hosted model adapters, while Local Demo uses a prepared local profile. Switching modes affects the next submission and shall not alter a running job. The accompanying explanation shall state which local features are unavailable and where latency or quality may differ. Any demonstration using sample data shall be clearly distinguished from a run that performs actual inference.

3.2 Hardware Interfaces

3.2.1 Supported Devices

Users interact through a keyboard, pointing device and display using browser interfaces. Supported speech recognition may use an optional microphone. The server requires CPU, memory, storage and network resources. A GPU can accelerate a qualified local model, but the standard profile does not require one.

3.2.2 Hardware Interactions

No custom hardware is controlled by the application. Users may refuse browser microphone permission and continue by typing. Rendering and inference shall stay within configured resource limits. If capacity is unavailable, the application shall report a controlled error or keep the work queued.

3.3 Software Interfaces

Table 3: Software interfaces

Component	Interface	Responsibility
Frontend and gateway	HTTP(S), JSON and media responses	Submit prompts/settings, check progress and stream/download artifacts.
Gateway and AI service	Internal REST/JSON	Analysis, clarification, storyboards, semantic checks, narration and critique.
MongoDB	Mongoose/database connection	Store prompt versions, storyboards, generation metadata and artifact references.
Redis/Bull	Queue interface	Schedule background jobs, limit concurrency and track operational status.
Knowledge index	FAISS plus metadata	Find relevant cinematography passages using a compatible embedding model.
Inference adapters	Hosted HTTPS or configured local runtime interface	Normalize model requests/results; report readiness, timeouts and model identity.
Remotion and FFmpeg	Validated composition input and controlled process invocation	Render the timeline, validate media and produce posters, captions and derivatives.

3.3.1 Principal Service Operations

The contracts below describe the main workflow operations. History and execution-profile fields belong to the same validated, versioned interface. Errors shall contain safe explanations and stable failure categories, excluding credentials and internal stack traces.

Table 4: Principal application service operations

Operation	Input	Result
POST /api/prompts	Prompt text; selected execution context	Prompt ID, analysis and applicable clarification questions.
POST /api/prompts/{id}/clarify	Question/answer pairs	A structured brief with the clarified prompt.
POST /api/storyboards	Prompt/version ID, style, render route and execution profile	Storyboard ID, shared world state and validated shots.

Continued on next page

Principal application service operations (continued)

Operation	**Input**	**Result**
POST /api/storyboards/{id}/generate	Accepted storyboard ID	A handle and status for the accepted job.
GET /api/jobs/{id}	Job identifier	Current stage, progress, shot outcomes and available artifacts.
POST /api/images	Prompt/version, image route and style	Image, effective prompt and artifact reference.
GET /api/storyboards/{id}	Storyboard identifier	Saved storyboard and associated result metadata.
GET /api/health	No content	Service health and readiness, identifying any affected dependencies.
History/version operations	Pagination, prompt/version identifiers	Stored inputs, revisions, comparisons and the corresponding generation records.

3.4 Communication Interfaces

Structured requests and intermediate records use JSON. Media responses carry the appropriate MIME type. The browser checks progress through bounded polling; a later push channel may use the same job-state contract. Application and model-service traffic crossing untrusted networks shall use HTTPS. Access to the database, queue and local runtime interfaces shall remain restricted to the deployment environment.

Each provider adapter shall declare its supported task, input/output format, limits and any seed or reference-image support. Timeouts, exhausted quotas and invalid responses shall become meaningful application errors. The operator shall configure the local runtime address; users shall not supply arbitrary runtime URLs. Requests shall preserve the chosen execution profile across all applicable stages.

4 Requirement Analysis

4.1 Functional Requirements

These requirements state the expected behaviour of VidCraft. Optional functions apply when enabled. Each requirement links to testing evidence through its ID in Section 6.6.

4.1.1 Prompt analysis and refinement

This module develops an initial idea into a reviewable brief. Linguistic scoring and focused questions address missing details while retaining the creator's intended subject, action and setting.

Table 5: Prompt analysis and refinement

ID	Requirement	Priority
FR-01	Prompt submission. The system shall accept nonblank text up to 4,000 characters, save the original and assign a persistent identifier. It shall reject invalid types and oversized text.	High
FR-02	Linguistic scoring. The analyzer shall assign integer scores (0–100) for subject clarity, action specificity, environment detail, visual richness and temporal coherence. The overall score shall equal their rounded arithmetic mean.	High
FR-03	Actionable feedback. The system shall identify configured gaps and contradictory visual terms, then suggest relevant improvements. Scores shall be described as linguistic indicators, not guarantees of media quality.	High
FR-04	Optional dictation. Supported browsers shall accept dictated prompts and clarification answers. Typing shall remain available when recognition is unsupported or microphone permission is denied.	Optional
FR-05	Bounded clarification. The system shall ask one focused question round for unresolved details, up to a configurable maximum of six questions, and validate each question–answer association.	High
FR-06	Creative brief. The system shall produce a reviewable brief from the original prompt and answers, retaining explicit subject/action/setting constraints and every version. Users shall be able to edit the brief before starting a new attempt.	High

4.1.2 Style and workflow configuration

This module saves visual and production choices for a confirmed brief. Style, mood, lighting and palette describe the intended appearance. The workflow, aspect ratio and media route determine how that appearance is produced.

The image workflow creates a single composition, while the storyboard workflow creates ordered shots that share a visual style. Before submission, the interface offers options supported by the route and execution profile.

Table 6: Style and workflow configuration

ID	Requirement	Priority
FR-07	Style propagation. The system shall save visual style, mood, lighting, palette and custom terms, carrying them into image prompts and storyboard world state. Explicit user choices shall override conflicting style inferred by the system.	High
FR-08	Output dimensions. The system shall offer aspect ratios supported by the chosen route only. Baseline video shall use 16:9 at 1920 by 1080. The system shall reject unsupported dimensions before generation starts.	High
FR-09	Workflow choice. The user shall choose single-image or storyboard generation. The image workflow shall skip shot decomposition, while the storyboard workflow shall follow an ordered plan of shots.	High
FR-10	Media route choice. The system shall use an explicitly selected compatible route, identify unavailable routes and disclose known costs or quotas. It shall never silently replace that route with a chargeable media provider.	High

Configuration example. A creator chooses a warm palette and soft lighting for a storyboard. These choices accompany shared scene information and each shot prompt. Explicit choices override any conflicting style inferred during planning. If settings change while generation is running, they apply to a later task; the active task keeps its recorded settings.

4.1.3 Multi-agent planning and cinematography

This module turns the confirmed brief into a production plan for review. The Screenwriter arranges the narrative and shared scene details. The Cinematographer uses the curated knowledge base to refine shot instructions. Validation and a bounded intent check govern transitions between these stages. The storyboard shows shot order, camera treatment and timing before committing resources to generation.

Table 7: Multi-agent planning and cinematography

ID	Requirement	Priority
FR-11	Narrative decomposition. The Screenwriter shall create three to five ordered shots, each with a nonempty description, camera instructions and finite positive duration, plus shared character and setting information. Invalid structured output shall be blocked from rendering.	High
FR-12	Retrieval-grounded planning. The Cinematographer shall retrieve relevant passages from the curated knowledge base, using a default top-k of three per shot, to refine camera/style instructions. Query and corpus embeddings shall share a matching model/vector space.	High
FR-13	Intent preservation. The system shall measure semantic similarity between the brief and planned narrative. At an initial threshold of 0.35, it shall allow at most two revisions after the first draft. It shall warn of low final scores and mark unavailable measurements unchecked.	High
FR-14	Scene continuity. The system shall combine shared character, setting and style details with each shot-specific action/camera instruction. Supported routes may use reference anchoring. Evaluation shall assess generated appearance separately from consistency of the supplied parameters.	High
FR-15	Storyboard review. Before media generation, the user shall be able to inspect shot order, descriptions, camera choices, duration and shared scene information. The interface shall let the user accept the plan or return to refinement.	High

4.1.4 Generation, quality control and delivery

Table 8: Generation, quality control and delivery

ID	Requirement	Priority
FR-16	Procedural rendering. The system shall map supported shot categories to wide, medium and close-up compositions. Unknown categories shall use a documented medium-shot fallback. Final assembly shall not require individual media URLs for completed procedural shots.	High
FR-17	Image synthesis. The system shall create a single image or storyboard shot image using the effective prompt and chosen style. It shall validate returned media and record the route and source-input references.	High
FR-18	Video synthesis. With a compatible video-generation route enabled, the system shall accept and validate generated clips and preserve their temporal content in assembly. It shall distinguish sequences of animated stills from clips containing synthesized motion.	Medium
FR-19	Background execution. The system shall queue generation asynchronously and return a job handle without waiting for rendering. Its initial resource-constrained profile shall allow one active generation worker.	High
FR-20	Duplicate protection. Concurrent start requests for the same active storyboard shall resolve to one effective generation job without repeating chargeable media work. A deliberate regeneration after that run shall create a separate attempt.	High
FR-21	Progress and recovery. The interface shall show the current stage, progress, shot outcomes and stage errors. After reload, it shall use the retained job handle to resume status lookup, stopping at terminal or expired states. A completed job with no video shall not appear as a successful video result.	High
FR-22	Controlled failure handling. The system shall bound request timeouts and retries, keeping completed shots if another fails. It shall report partial/no-output results explicitly when applicable. External-service failures shall preserve inputs and successful artifacts.	High
FR-23	Visual feedback. When enabled for compatible generated visuals, the Critic shall record a verdict and reason. By default, it shall request at most two regenerations per shot, retaining the final usable asset when exhausted. Unavailable critique shall be marked unchecked.	Optional
FR-24	Narration. When enabled, the system shall create narration beats for usable shots, synthesize speech and set timing from measured beat durations. Failed speech shall leave a recorded silent interval or silent output while preserving the visual assets.	Optional

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Generation, quality control and delivery (continued)

ID	Requirement	Priority
FR-25	Common media timeline. The system shall assemble usable shots in order, using the same inclusion rule for visuals, narration and captions. At 30 FPS, each included shot shall use max(1, round(duration in seconds times 30)) frames. Failed or held shots shall add no timeline interval.	High
FR-26	Post-processing. After assembly, the system shall create a poster, WebVTT captions and a web-compatible MP4. It may create a separate burned-in subtitle copy. Optional derivative failure shall preserve the master.	High
FR-27	Colour treatment. The system should provide a named, reproducible colour-grading preset for a derived video while retaining the master and preserving dimensions, duration and audio synchronization.	Medium
FR-28	Preview and download. Users shall be able to preview available media and download the master, images, captions, poster and enabled variants. The interface shall identify missing or failed artifacts without preventing valid downloads.	High
FR-29	Regeneration records. Each new generation attempt shall keep its own input, settings, mode and output attribution. Earlier retained results shall remain linked to the attempt that originally produced them.	Medium

4.1.5 History and version comparison

This module connects each prompt, revision and generation attempt with its inputs and settings. These records help creators trace results and evaluators examine the team's prompt-refinement workflow.

Within the permitted deployment or session scope, paginated history shows timestamps, status, execution mode and links to retained results.

Table 9: History and version comparison

ID	Requirement	Priority
FR-30	Saved history. The system shall display paginated prompt and generation history within the permitted deployment/session scope, showing timestamps, status, selected mode and links to results that remain available.	Medium
FR-31	Prompt versions. The system shall save original, clarified and user-edited versions. Selected versions shall be compared through text differences and subject/action/setting/style attribution. Semantic comparisons shall name their encoder and reject incompatible vector spaces.	Medium

Version interpretation. Comparison separates original, clarified and edited text, grouping changes by subject, action, setting and style. Semantic similarity provides additional context when the embedding spaces match. It does not establish improved media quality.

Illustrative workflow. A creator expands "a cyclist on a road" into a mountain-road scene at sunrise and later adjusts the lighting. History keeps these revisions and their outputs distinct, helping the creator choose between further editing and regenerating the saved plan.

Record and artifact availability. Media files may expire before their records are removed. History identifies unavailable artifacts while retaining the attempt's identity. Earlier results keep their association with their original settings.

4.1.6 Hybrid inference and project evaluation

Table 10: Hybrid inference and project evaluation

ID	Requirement	Priority
FR-32	Execution-mode control. A compact Standard / Local Demo control shall default to Standard and display mode and readiness. The interface shall explain local performance limits without obscuring workflow.	High
FR-33	Consistent mode routing. Each task shall keep a server-validated profile covering all enabled language tasks. Changes to the UI mode shall affect subsequent tasks only.	High
FR-34	Local model readiness. The local profile shall verify runtime, installed model, structured-output support and limits. Local and Standard outputs shall meet identical validation contracts.	High
FR-35	Local retrieval. Local Demo shall use local embeddings and a matching index for retrieval and intent checking. Unavailable resources shall be marked setup-required or unchecked; hosted embedding calls shall not be used.	High
FR-36	Local capability profile. Local Demo shall support local language processing, embeddings and procedural rendering. Visual generation, critique, speech and dictation shall be disabled unless supported locally. Hosted processing shall require explicit mode selection.	High
FR-37	Model provenance. Records shall name the model/runtime, mode, quantization and adapter version. Fine-tuning claims shall require training artifacts; pretrained or quantized models alone shall not qualify.	High
FR-38	Adaptation experiment. If fine-tuning is performed, the team shall record the base model, task, data provenance, training settings and checkpoint/adapter. It shall compare the adaptation with the unchanged base on disjoint held-out prompts.	Optional
FR-39	Configuration and readiness. The operator shall configure models/routes, retry limits, feature flags and corpus/index preparation. Health reports shall distinguish process availability from storage, queue, model and renderer readiness.	High
FR-40	Comparative evaluation. The project shall assess refinement and baseline-versus-multi-agent planning using 50 frozen prompts, and compare Standard/Local Demo on a declared paired set. Reports shall include failures, actual sample counts, model versions, timings and quality measures without assuming improvement.	High

4.2 Use-Case Descriptions

The Creator is the main actor. Operator and Evaluator denote project roles, not a separate public account-management subsystem. Model Services is a secondary actor: the hosted services or local runtime that the system calls while serving a use case. Common failure handling retains typed input, the confirmed brief and stored artifacts. Each use case describes the expected sequence, alternative paths and exceptional conditions.

4.2.1 Use-Case Diagram

Figure 6 places the twelve use cases inside a single system boundary. The primary actor is on the left and the secondary and supporting actors are on the right. Two relationships connect use cases:

- UC-02 «extend» UC-01: clarification is added to prompt analysis only when unresolved details remain.
- UC-09 «include» UC-06: regenerating from a saved plan always performs storyboard media generation.

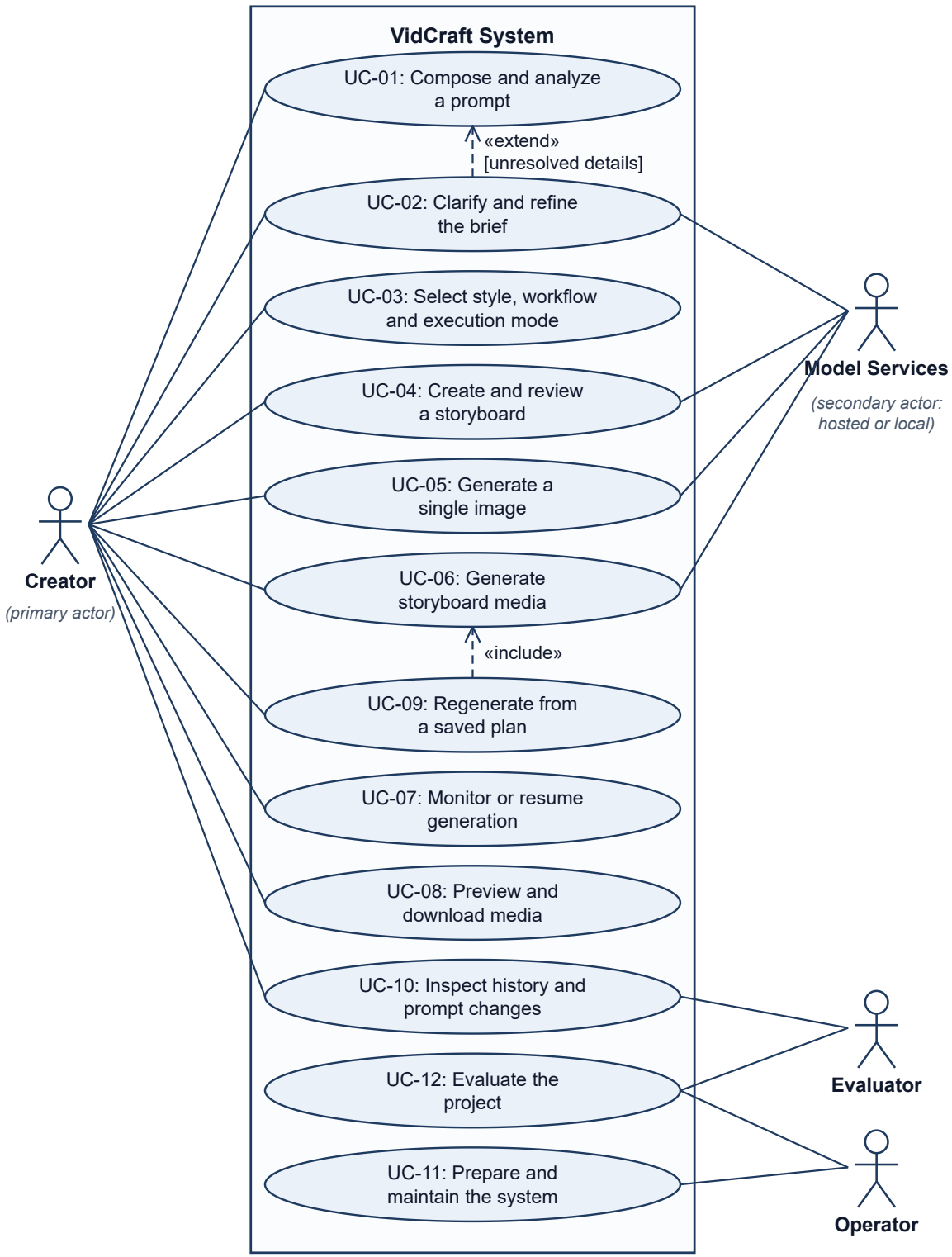


Figure 6: Use-case diagram of VidCraft

4.2.2 UC-01: Compose and analyze a prompt

Table 11: Use case UC-01: Compose and analyze a prompt

Attribute	**Description**
Actors	Creator
Preconditions	The studio and analyzer are available.
Trigger	The user chooses Analyze.
Basic flow	1. Type or optionally dictate a prompt. 2. Validate the input. 3. Save the original and calculate scores. 4. Show dimension scores, flags and suggestions. 5. Offer clarification if needed.
Alternative flow	Detailed prompts proceed directly to settings; typing remains available if dictation is unsupported.
Postconditions	A prompt identifier and analysis are available.
Exceptions	Blank/oversized input, refused microphone permission or analyzer failure.
Requirements	FR-01–04

Interaction detail. Scores direct the creator to descriptions that may need more detail. A prompt might clearly name its subject but omit the action or surroundings. The accompanying feedback links these omissions to possible refinements. A high score indicates descriptive completeness; it cannot promise that the generated image or video will match the intended quality.

4.2.3 UC-02: Clarify and refine the brief

Table 12: Use case UC-02: Clarify and refine the brief

Attribute	**Description**
Actors	Creator (primary); Model Services (secondary)
Preconditions	An analyzed prompt still contains unresolved details.
Trigger	The user requests clarification.
Basic flow	1. Show the bounded question set. 2. Record each corresponding answer. 3. Form a clarified brief. 4. Present the result. 5. Accept or edit it and save its version.
Alternative flow	When clarification is unnecessary, use the original prompt; save edits as new versions.
Postconditions	The reviewed creative brief is ready.
Exceptions	Unanswered questions, invalid model structure or inference timeout; supplied answers are retained.
Requirements	FR-05–06, FR-31

Interaction detail. Questions address details still missing from the submitted idea. Keeping each answer with its question helps the revised brief reflect what the creator actually supplied. The creator reviews the combined text before planning and can correct an interpretation that changes the intended scene. Original and revised versions stay distinct for subsequent comparison.

4.2.4 UC-03: Select style, workflow and execution mode

Table 13: Use case UC-03: Select style, workflow and execution mode

Attribute	**Description**
Actors	Creator
Preconditions	A usable prompt and known capabilities are available.
Trigger	The user opens the generation settings.
Basic flow	1. Choose image or storyboard. 2. Set a style and supported aspect ratio. 3. Choose Standard or Local Demo. 4. Select an available media route. 5. Validate and save settings for the request.
Alternative flow	Local Demo is disabled if its runtime is unavailable; settings changed during processing affect the next request.
Postconditions	The next task retains a valid settings/profile snapshot.
Exceptions	An incompatible ratio/route, an unavailable model or an exhausted budget.
Requirements	FR-07–10, FR-32–36

Interaction detail. The three selections serve related purposes: workflow sets the form of the output, execution mode sets the processing profile, and media route selects a compatible generation capability. The saved configuration keeps the combination submitted for that task. Local Demo must pass its readiness check, and its available functions may be fewer than those offered through hosted services.

4.2.5 UC-04: Create and review a storyboard

Table 14: Use case UC-04: Create and review a storyboard

Attribute	**Description**
Actors	Creator (primary); Model Services (secondary)
Preconditions	A confirmed brief and compatible settings exist.
Trigger	The user requests a storyboard.
Basic flow	1. The Screenwriter forms ordered shots and shared world state. 2. The Cinematographer retrieves relevant techniques. 3. Validate the plan. 4. Run the bounded intent check. 5. Show shots for acceptance or refinement.
Alternative flow	Low similarity permits only the specified revisions; missing measurements remain marked unchecked.
Postconditions	A structured plan is stored and reviewed before rendering.
Exceptions	Malformed plan, incompatible index or model failure.
Requirements	FR-11–15

Review focus. The creator checks that the shot sequence expresses the intended action and retains the shared characters, setting and visual style. Camera instructions and durations explain how the sequence will unfold. Any similarity warning informs this review but leaves the final judgment with the creator. Accepting the storyboard confirms the plan for this generation attempt.

Planning responsibility. Shared scene details support continuity, and retrieved techniques guide camera choices. Team-developed orchestration combines these representations and validation into a reviewable plan. Generated appearance is checked separately after rendering.

4.2.6 UC-05: Generate a single image

Table 15: Use case UC-05: Generate a single image

Attribute	Description
Actors	Creator (primary); Model Services (secondary)
Preconditions	A confirmed prompt and image-capable route are selected.
Trigger	The user chooses Generate image.
Basic flow	1. Combine the prompt with its style. 2. Call the chosen adapter. 3. Validate the returned image. 4. Save the result and provenance. 5. Provide preview and download.
Alternative flow	After failure, the user may change settings and retry; the workflow does not decompose a storyboard.
Postconditions	One attributable image or an explicit failure outcome is available.
Exceptions	Timeout, unsupported format, provider limit or storage failure.
Requirements	FR-09–10, FR-17, FR-28

Result interpretation. The image request uses the reviewed description together with the selected style. Its result retains those inputs and the actual generation route for later review. If that route fails or returns unusable media, the record remains a failed image attempt. The system neither turns it into a storyboard nor automatically selects another chargeable route.

Review focus. In the preview, the creator checks the subject, action, setting and chosen style. Recorded input details help guide a deliberate revision if the image departs from the brief.

4.2.7 UC-06: Generate storyboard media

Table 16: Use case UC-06: Generate storyboard media

Attribute	**Description**
Actors	Creator (primary); Model Services (secondary)
Preconditions	An accepted storyboard exists and required components are ready.
Trigger	The user starts generation.
Basic flow	1. Deduplicate the request and issue a job handle. 2. Process shots, keeping their outcomes. 3. Run optional bounded critique. 4. Prepare narration for usable shots when enabled. 5. Assemble and post-process the media.
Alternative flow	Failed shots are consistently excluded; failures in optional stages preserve other valid results.
Postconditions	The run records complete, partial, no-output or failed results and available artifacts.
Exceptions	A model failure, exhausted resources, queue interruption or rendering error.
Requirements	FR-16–27

Processing detail. Before assembly, the system records an outcome for each shot. One inclusion rule governs visuals, narration and captions so that failed shots do not create gaps or mismatched segments. Optional critique and narration remain within their configured limits. The final status states whether the run produced complete media, partial output or no usable result.

4.2.8 UC-07: Monitor or resume generation

Table 17: Use case UC-07: Monitor or resume generation

Attribute	**Description**
Actors	Creator
Preconditions	A job handle is retained.
Trigger	The user checks progress or reloads.
Basic flow	1. Fetch the job state. 2. Show its stage and shot outcomes. 3. Refresh according to the polling policy. 4. Reuse the saved handle after reload. 5. Stop at terminal state and display artifacts.
Alternative flow	Temporary disconnection triggers bounded backoff; an expired job displays recovery guidance.
Postconditions	The user sees the actual state and available results.
Exceptions	Unreachable backend, expired handle or interrupted worker.
Requirements	FR-19–22

Recovery detail. After reload, the retained handle reconnects the interface to the existing job without resubmitting generation. Progress reflects the reported stage and actual shot outcomes; starting a stage does not establish that it succeeded. Polling stops when the job becomes terminal or expires, and the interface shows the available results or explains the applicable recovery action.

Status interpretation. Finishing a queue operation does not necessarily produce usable media. Before reporting successful delivery, the result view checks available artifacts and retains warnings for partially completed stages.

4.2.9 UC-08: Preview and download media

Table 18: Use case UC-08: Preview and download media

Attribute	Description
Actors	Creator
Preconditions	At least one artifact is available.
Trigger	The user opens a result.
Basic flow	1. Display the available preview. 2. Provide caption/poster/variant links. 3. Play media using controls. 4. Download the chosen artifact. 5. Show warnings from any partially completed stages.
Alternative flow	An unavailable optional derivative leaves the master or source image accessible.
Postconditions	The chosen valid artifact is retrieved.
Exceptions	Expired file, unsupported browser format or network failure.
Requirements	FR-24–28

Delivery detail. The result view separates the master, supporting files and optional derivatives. A creator can therefore retrieve a valid video even if its poster, subtitle copy or another optional output is missing. Warnings remain attached to partial results. Downloading an available artifact should not give the impression that every stage of the request finished successfully.

Artifact selection. The master, captions, poster and enabled variants each have a clear identity. Creators can choose the deliverable they need and see which optional outputs the run did not produce.

4.2.10 UC-09: Regenerate from a saved plan

Table 19: Use case UC-09: Regenerate from a saved plan

Attribute	Description
Actors	Creator
Preconditions	The earlier attempt is terminal and its plan is retained.
Trigger	The user chooses Regenerate.
Basic flow	1. Inspect the source plan and current settings. 2. Record a new attempt with a fixed profile. 3. Queue generation. 4. Show its progress. 5. Keep earlier outputs linked to their original attempts.
Alternative flow	If an attempt is active, return its existing handle without duplicating generation.
Postconditions	A distinct, separately identifiable attempt is saved.
Exceptions	Unavailable route or exhausted resources; earlier results remain accessible.
Requirements	FR-20, FR-29

Attempt separation. Regeneration uses the reviewed source plan but records new execution settings and a fixed profile. Retained earlier outputs keep their original input links, allowing attempts to be compared without overwriting history. Duplicate protection prevents repeated work during an active attempt. A deliberate later regeneration receives its own record.

Comparison focus. Separate records show whether settings changed between attempts or the generated results simply varied. The earlier input and output give the creator context for deciding what to submit next.

4.2.11 UC-10: Inspect history and prompt changes

Table 20: Use case UC-10: Inspect history and prompt changes

Attribute	**Description**
Actors	Creator; Evaluator
Preconditions	Stored prompts and their versions are available.
Trigger	The user opens History.
Basic flow	1. Browse the paginated entries. 2. Choose a prompt and versions. 3. Show text differences. 4. Present semantic attribution and compatible similarity. 5. Open the linked outputs.
Alternative flow	Expired media, uncertain attribution and incompatible embeddings are explicitly identified.
Postconditions	The result can be traced to its source version and settings.
Exceptions	A missing record, incompatible encoder or unavailable artifact.
Requirements	FR-30–31

Comparison detail. Text differences identify the edits between two versions, while subject, action, setting and style attribution explains how the brief changed. A compatible semantic comparison supplies additional context. Missing measurements are reported explicitly. An expired artifact is also distinguished from a missing prompt record, so users can see where the available evidence stops and where comparison remains possible.

Traceability focus. Each selected result points back to its prompt version and saved configuration. Creators use this link during revision, and evaluators use it when reviewing the prompt-refinement process.

4.2.12 UC-11: Prepare and maintain the system

Table 21: Use case UC-11: Prepare and maintain the system

Attribute	**Description**
Actors	Operator
Preconditions	The operator has trusted configuration access.
Trigger	Setup or maintenance is required.
Basic flow	1. Inspect service readiness. 2. Set permitted routes and limits. 3. Prepare the model and matching index. 4. Validate a small task. 5. Carry out controlled backup or cleanup as needed.
Alternative flow	If an optional component is missing, only its dependent feature is disabled.
Postconditions	Supported profiles and retained data can be used.
Exceptions	Invalid settings, incompatible model/index, insufficient storage or an unsuccessful backup.
Requirements	FR-34–39, OR-01–03

Readiness detail. A running process alone does not establish readiness. The chosen profile needs usable storage, a queue, the required model resources and a renderer that work together. A small validation task checks these dependencies before a demonstration. Preparing the corpus alongside its embedding model ensures that indexed passages and retrieval queries belong to a compatible vector space.

Demonstration preparation. Before a session, the operator checks the intended mode and available functions. Missing local models and incompatible indexes remain visible readiness problems until the corresponding resources have been prepared.

4.2.13 UC-12: Evaluate the project

Table 22: Use case UC-12: Evaluate the project

Attribute	**Description**
Actors	Evaluator; Operator
Preconditions	The inputs are frozen, models prepared and protocol declared.
Trigger	The evaluator starts an evaluation run.
Basic flow	1. Define baseline and comparison conditions. 2. Process identical controlled inputs. 3. Save outputs, timings, provenance and failures. 4. Calculate specified metrics and human ratings. 5. Present findings and limitations.
Alternative flow	Fine-tuning adds a separate base-versus-adapted comparison; interrupted runs state the actual sample size.
Postconditions	Conclusions about the developed workflow have reproducible supporting evidence.
Exceptions	Quota interruption, a model change, data leakage or unavailable outputs.
Requirements	FR-37–40

Evidence interpretation. Evaluation uses controlled inputs under declared conditions and keeps failed attempts as well as successful outputs. Model and runtime records describe how each run was processed. Timing and quality measures cover different aspects of performance. Findings about refinement, planning and local execution must follow the observations. An adaptation experiment reports its base-model comparison and held-out evaluation evidence.

4.3 Nonfunctional Requirements

These values set acceptance targets for the stated reference configuration, not completed benchmarks. Measurements must distinguish cold and warm runs and record models, input sizes and enabled features. The requirements cover a controlled academic deployment. Public-service obligations apply if such a deployment is introduced.

Table 23: Nonfunctional requirements

ID	Requirement and acceptance target	Priority
NFR-01	Responsiveness. The interface shall display loading or validation feedback within 200 ms of an ordinary action in the reference browser.	High
NFR-02	Service latency. With healthy local dependencies, status requests shall achieve p95 of 1 second and job admission p95 of 2 seconds across at least 20 measurements, excluding model/render execution.	High
NFR-03	Bounded execution. Each model/render stage shall define a finite timeout and retry cap. Initial limits are 60 seconds per hosted language call, 180 seconds per local language call and 900 seconds per generation job; deployment tests shall verify their suitability.	High
NFR-04	Resource efficiency. The reference profile shall allow one active generation and one local inference at a time, preserve a configured free-disk reserve and reject unavailable capacity without losing saved work.	High
NFR-05	Reliability. Failure of a shot or optional stage shall preserve successful assets. The system shall reconcile interrupted work before a retry that could cause another external charge.	High
NFR-06	Usability. At least four of five representative first-time users should complete prompt-to-download within ten minutes of active interaction, excluding generation waits, with only the supplied guidance.	Medium
NFR-07	Accessibility. The workflow shall allow keyboard operation, visible focus, labeled controls and status feedback independent of colour. Layouts shall remain usable at 360, 768 and 1440 CSS-pixel widths; ordinary text has a contrast target of 4.5:1.	High
NFR-08	Security. Credentials shall be excluded from client bundles and public logs. Inputs, paths and model responses shall be validated, and generated text shall never execute as code. Authentication/authorization and HTTPS shall precede exposure of a shared deployment.	High
NFR-09	Privacy. The interface shall explain enabled external processing and microphone use. Diagnostics shall exclude secrets and unnecessary prompt content. Local mode shall follow its capability profile without invoking hosted inference.	High

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Nonfunctional requirements (continued)

ID	Requirement and acceptance target	Priority
NFR-10	Maintainability. Analysis, orchestration, model adapters and rendering shall use explicit interfaces and centralized configuration. Replacing an adapter shall leave unrelated workflow contracts unchanged.	Medium
NFR-11	Portability. Setup instructions shall cover the selected Windows/WSL or Linux environment, including dependency manifests, safe configuration examples and repeatable steps for preparing models and indexes.	Medium
NFR-12	Media compatibility. Baseline MP4 output shall decode at 1080p/30 FPS in the tested desktop browsers. Captions shall match the final timeline within one frame; narration shall target alignment within 100 ms of declared beat boundaries.	High
NFR-13	Reproducibility. Evaluation records shall retain dataset/model/profile identity, settings, outputs, failures and timing. Changed models shall define separate experimental conditions unless pooling is justified.	High
NFR-14	Budget control. Chargeable calls and retries shall be recorded against an operator-defined call or spend cap. Unknown charges shall not be reported as zero; changing to a paid route shall require selection.	High
NFR-15	Scalability. The queue shall accept at least three test jobs, keep status requests responsive and process jobs at the configured concurrency without mixing state or artifacts. Additional workers may be added without changing public contracts.	Medium
NFR-16	Recoverability. The operator shall back up retained assessment data at least daily during active use. A restore test shall recover consistent records and included media, identifying incomplete jobs without automatically repeating paid work.	Medium

5 Other Requirements

5.1 Database Requirements

MongoDB shall link each prompt to its versions, storyboards, generation attempts and artifact references. Redis manages queue execution; persistent history remains in the database. Generated media shall reside in a controlled directory or a configured artifact store. The cinematography index shall retain its corpus and encoder metadata. Prompt-comparison embeddings shall remain distinguishable from cinematography retrieval vectors.

Table 24: Logical data dictionary

Entity	**Principal information**	**Key relationships / constraints**
Prompt	Identifier, original text and timestamps.	The original wording is preserved.
Prompt version	Version/parent ID, prompt ID, clarified or edited text, brief and optional embedding reference.	Belongs to one prompt; earlier versions remain intact.
Analysis	Five scores, overall score, flags and suggestions.	Belongs to one prompt version; scores are 0 to 100.
Storyboard	Source version, shared world state and ordered shots.	References one source version and plans three to five shots.
World state	Characters, setting, style tokens and optional reference image.	Supplies common context to the storyboard.
Shot	ID, description, camera, positive duration, route, status, asset and critique.	ID is unique within its storyboard; procedural shots may lack separate assets.
Narration beat	Action, spoken text, measured/silent duration, voice and audio reference.	Ordered within a usable shot and recorded by one generation attempt.
Generation attempt	Source, settings, execution profile, job ID, stage/outcome, timings and errors.	Separates runs and records the actual model/route.
Artifact	Attempt ID, type, URI, MIME type, dimensions/duration and retention state.	References its source/master and indicates unavailable files.
Technique passage / index	Stable ID, text, category, technique, tags, encoder and index version.	Uses unique passage IDs and a compatible vector space.
Evaluation record	Attempt and condition IDs, model settings, output references, measures and failures.	References the frozen dataset and evaluation protocol.

5.2 Logical Data Model

Figure 7 shows the logical entity–relationship diagram in crow's-foot notation. It covers the eleven entities of the data dictionary, arranged on a regular grid so that every relationship line is straight. The model is logical: an entity may be stored as an embedded MongoDB document rather than as a separate collection. A generation attempt for a single image has no storyboard, so an attempt refers to zero or one storyboard. A derivative artifact, such as a captioned copy, refers to its master artifact through `master_id`.

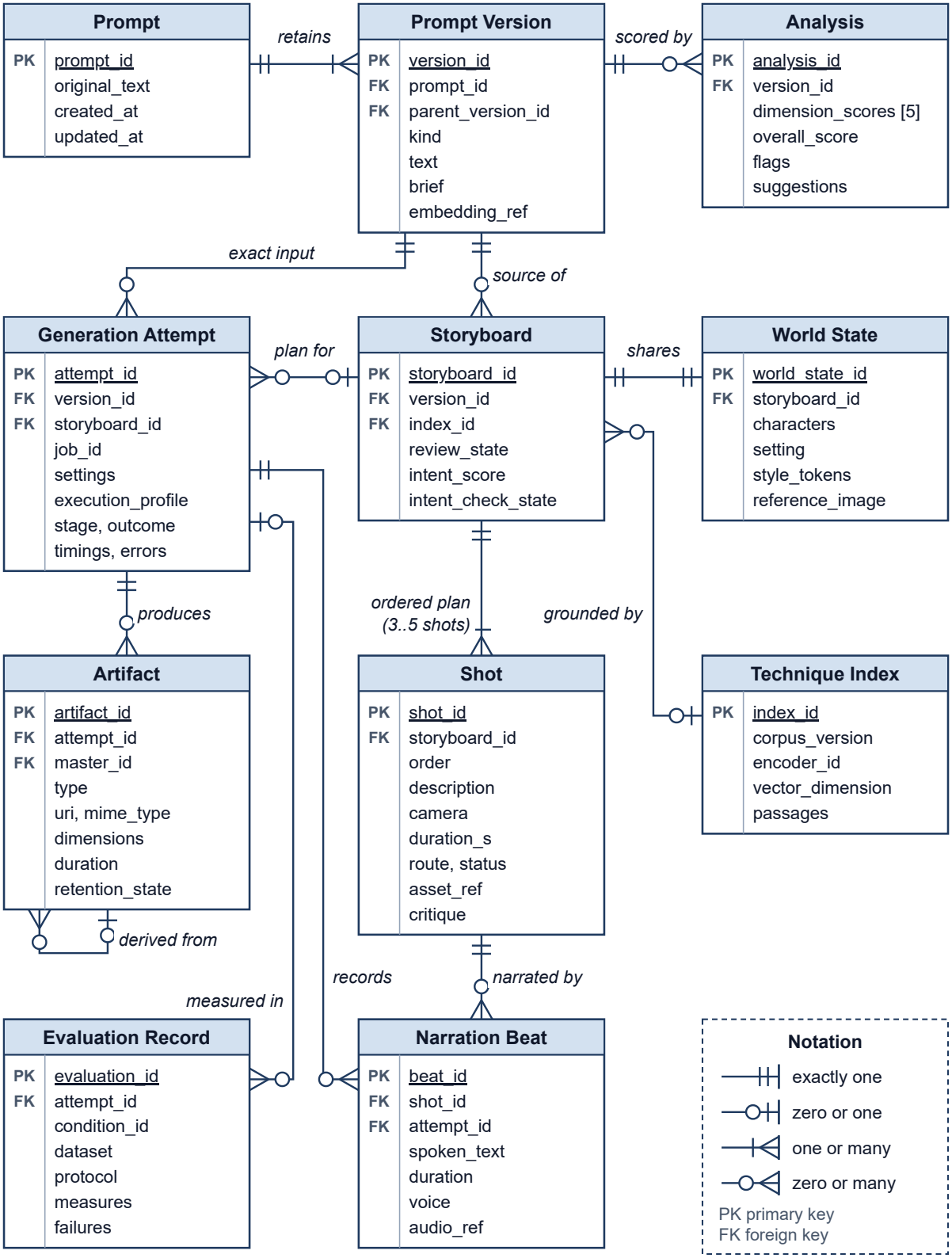


Figure 7: Logical entity–relationship diagram (crow's-foot notation)

Table 25: Data and supporting requirements

ID	Requirement	Priority
OR-01	Data integrity. Identifiers, references, finite durations and permitted states shall be validated before storage or rendering. Artifacts shall become available only after file finalization and validation.	High
OR-02	Version and run attribution. Retained outputs shall identify input version, settings and attempt. Later model/profile changes shall leave provenance unchanged.	High
OR-03	Retention and cleanup. The operator shall document retention periods. Cleanup shall remove only eligible managed records/files, protect active jobs and selected assessment artifacts, and mark expired media in history.	Medium
OR-04	Ethical use and provenance. The team shall document sources/licenses for software, models, corpora and datasets, preserve required attribution, follow provider restrictions and report experiments and generated content accurately.	High
OR-05	Documentation. The team shall maintain setup/user guides, component/API descriptions, diagrams, test records and an evaluation report reflecting the delivered version.	Medium

5.3 Ethical and Legal Requirements

Users shall be told that outputs are generated media and need review before publication. The team shall use appropriately sourced components and datasets, document attribution and avoid unsupported ownership or originality claims about foundation models. Experiment records shall distinguish pretrained, quantized and fine-tuned models. Any fine-tuning shall use separately recorded training, validation and test data, keeping assessment prompts out of training.

5.4 Privacy Requirements

The system shall retain only information needed for authoring, generation, troubleshooting and agreed evaluation. Microphone access shall remain optional. Users shall be informed when an enabled service receives their prompt or media. Local-mode checks shall confirm that disabled hosted tasks are not called. Deletion statements shall cover application-controlled storage only; the selected service determines retention on the provider side.

The initial system runs in a trusted local/laboratory environment. If access is shared, only authorized users shall be able to access the relevant prompts, jobs and media. Knowing a record identifier shall not be enough to obtain access. Secrets, private network addresses and full stack traces shall not be exposed to ordinary users.

5.5 System Documentation

Deliverables shall include the SRS, design diagrams, installation and user guides, source and dependency information, and test and evaluation reports. Operator instructions shall explain startup, readiness checks, local models, index rebuilding, output types, cleanup and backup. Each test record shall name the tested version and actual outcome. No test pass or benchmark result is presumed by this specification.

5.6 Business Rules and Processing States

1. Standard inference is selected by default. A user selects Local Demo explicitly, and its profile remains fixed for the attempt.
2. Submission is allowed only for compatible combinations of workflow, media route and dimensions.
3. Refining a prompt or generating another result shall preserve its original text and retained versions.
4. A plan may have an initial draft and at most two revisions. Optional visual critique allows at most two regenerations per shot by default.
5. Retrieval and semantic comparison require matching encoder/index identities. An unavailable quality check shall not count as a successful measurement.
6. The final visual, audio and caption timeline shall consistently exclude failed or held shots.
7. Queue completion alone does not establish that an artifact exists. Results shall distinguish full success, partial output, no output and infrastructure failure.
8. After a local model failure, hosted processing shall require an explicit user choice before retrying the task.

Shots use the states pending, processing, completed, failed and on hold; queue states are waiting, active, delayed, completed and failed. Appendix B gives the corresponding state-transition diagrams (Figures 11 and 12).

6 System Testing

6.1 Test Cases

Each case states its input or condition and expected outcome. Controlled fixtures check processing rules and failure handling; real model and media runs check integration. These cases form a test plan rather than a completed test report.

Table 26: System test cases

ID	Test	Input / condition	Expected result
TC-01	Prompt validation	Normal English prompt; blank/non-string input; 4,000 and 4,001 characters.	Accept and preserve only valid, bounded input; reject invalid input with a clear error.
TC-02	Scores and contradictions	Fixed prompts missing subject/action/setting, with configured contradictory terms.	Return five bounded scores and their correct mean, relevant flags/suggestions and repeatable analysis.
TC-03	Dictation alternatives	Supported browser, refused permission, unsupported recognition; keyboard/pointer release.	Stop transcription correctly and keep typed input usable in every alternative.
TC-04	Clarified brief	Vague prompt, six-question cap, aligned/misaligned answers and a user revision.	Use one round of at most six questions, map answers correctly, preserve explicit intent and save the revised brief as a version.
TC-05	Style and dimensions	Different style/palette selections; supported and unsupported ratios.	Pass selections to image and storyboard inputs, reject unsupported ratios and render the selected dimensions.
TC-06	Workflow and route	Single image versus storyboard; incompatible or unavailable routes.	Follow the selected branch, omit shot planning for images and prevent silent paid substitution.
TC-07	Storyboard contract and review	Valid three/five-shot plans; invalid counts, empty descriptions or nonfinite durations.	Reject invalid plans and display valid ordered shots/world state for review before rendering.
TC-08	Retrieval compatibility	Controlled technique corpus with matching/mismatching encoder indexes.	Include relevant passages in planning and detect incompatible vector spaces before querying them.

Continued on next page

System test cases (continued)

ID	Test	Input / condition	Expected result
TC-09	Intent retry limit	Forced low scores and an unavailable embedding measurement.	Produce at most three drafts, warn when retries are exhausted and mark unavailable measurement as unchecked.
TC-10	Continuity inputs	Three shots with shared character/setting but different framing.	Pass shared context to each shot while varying action/camera; assess visual continuity separately.
TC-11	Procedural rendering	Wide, medium, close-up and unknown-camera inputs without shot image URLs.	Select supported compositions, apply the default for unknown cameras and render a valid final video.
TC-12	Image generation	One authorized real image route, plus invalid-media and failure fixtures.	Deliver a decodable image with source metadata or identify the failure while preserving input.
TC-13	Generative video	One real enabled video-model clip and final assembly.	Preserve temporal frames through playback/assembly; do not classify still-image animation as synthesized motion.
TC-14	Queue and concurrency	Five concurrent starts for one plan and three separately queued plans.	Create one effective shared-plan job, isolate other plan states and respect configured concurrency.
TC-15	Progress and reload	Reload during generation, temporary disconnect, expired job and worker completion without a video.	Resume retained-handle lookup, terminate polling correctly and report the actual artifact outcome.
TC-16	Failures and limits	Provider timeout/429, one/all failed shots, memory/disk pressure and a small call cap.	Stop within configured limits, respect caps, retain saved inputs/assets and distinguish partial from no-output results.
TC-17	Critic loop	Repeated negative verdicts, disabled critic and a critic outage.	Permit at most two regenerations, retain the final usable asset and distinguish disabled/unchecked from passed.
TC-18	Timeline and speech	Successful/failed shots, measured beats, one TTS failure and a known timing fixture.	Use included shots only for frames/audio/captions, mark silent recovery and meet timing and media-format targets.

Continued on next page

System test cases (continued)

ID	Test	Input / condition	Expected result
TC-19	Post-processing	Valid master; poster/VTT; optional burn-in or colour preset; forced derivative failure.	Produce usable derivatives, preserve the master and timing/dimensions, and report each failed stage independently.
TC-20	Result delivery	Each available artifact type and an expired/missing file.	Decode previews/downloads correctly and identify unavailable artifacts while keeping valid results visible.
TC-21	History and attribution	Multiple versions/runs, records beyond one page and incompatible comparison encoders.	Provide stable pagination, correct text/semantic changes, distinct attempts and preserved provenance.
TC-22	Mode isolation	Two tasks in different modes; toggle changed during an active run.	Start in Standard by default; retain each task's profile without changing the active run's mode.
TC-23	Readiness	Stopped queue/model/renderer, malformed local JSON and absent local index.	Mark affected operations unavailable, stop safely on schema violations and explain model/index preparation.
TC-24	Real local execution	Prepared small model/index, procedural route and monitored network requests.	Run actual local inference without hosted calls, disable unsupported tasks and retain model/adapter identity.
TC-25	Research integrity	Frozen 50-prompt set, paired conditions, recorded failures and optional adapter experiment.	Report correct sample counts/provenance, disjoint training/test splits and reproducible metrics; exclude simulated or unsupported training claims.
TC-26	Responsiveness	Twenty status/admission requests on the declared machine, plus ordinary UI actions.	Meet NFR targets for measured UI feedback and p95 service times; separate cold/provider delays in reporting.
TC-27	Usability and access	Five first-time users, keyboard-only operation, 360/768/1440 widths and contrast checks.	Record completion/time, provide accessible controls and usable responsive views, and retain failed attempts in results.
TC-28	Integrity and security	Invalid references/paths, hostile generated strings, dummy secrets in errors and cross-scope requests.	Prevent untrusted execution/storage escape, reject invalid data and enforce safe errors and appropriate access restrictions.

Continued on next page

System test cases (continued)

ID	**Test**	**Input / condition**	**Expected result**
TC-29	Maintenance and setup	Isolated setup using the documentation and a conforming replacement adapter.	Initialize services with stated prerequisites, preserve task contracts and confirm documentation matches behaviour.
TC-30	Backup and cleanup	Backup/restore a small dataset; remove expired data during an active job.	Restore matching records/files, protect active/retained artifacts, restrict cleanup to managed roots and avoid replaying paid work.

6.2 Testing Methodologies

Black-box testing examines inputs and visible results for generation, history and recovery. **White-box testing** examines implemented scoring rules, validation branches, agent routing, retry limits and frame calculations. **Gray-box testing** uses request contracts and stored records to check component interactions. **Regression testing** repeats the relevant checks when models, adapters or workflow logic change. **Acceptance testing** demonstrates agreed requirements with representative users and actual outputs.

6.3 Types of Testing and Tools

Table 27: Testing types and suitable tools

Type	**Project-specific purpose**	**Tools / method**
Unit	Scoring, contradiction rules, structured-output checks, vector compatibility and frame/caption calculations.	pytest; Node.js built-in test runner.
Integration	Gateway/AI/database/queue interaction, provider contracts and execution-profile transfer.	HTTP test fixtures; Postman or equivalent REST client; FastAPI API documentation.
System / browser	Authoring-to-download workflow, reload, mode selection and error displays.	Browser developer tools; optional Playwright; manual checks.
Performance	Service latency, render duration, local memory and bounded concurrency.	Timestamped test harness; OS process monitoring; optional Locust.
Security / privacy	Input/path checks, safe errors, secret handling, scoped access and network isolation in local mode.	Controlled negative fixtures; browser/network inspection.
Media validation	Container/codec details, frame counts, audio alignment and caption timing.	FFmpeg/ffprobe; browser playback; known timing fixtures.
User acceptance	Feedback clarity, workflow completion and the local demonstration.	Recorded tasks with representative users.
Regression	Changes in contracts, timeline processing and failure recovery.	Repeatable pytest/Node/browser checks relevant to the change.

6.4 Evaluation of the Project Contribution

The evaluation shall assess what the developed workflow contributes beyond the underlying model. A single enhancement step forms the baseline; the multi-agent condition follows the specified clarification, retrieval and planning process. Both conditions shall use the same fixed set of 50 single-beat and multi-beat prompts under controlled settings. For interrupted runs, reports shall state the completed pair count.

The comparison of standard and local processing shall measure structured-output validity, retention of user constraints, latency, failure rate, resource use and narrative quality. Wherever feasible, both conditions should use the same renderer and corpus to isolate language-planning effects. Changes in the visual-generation route shall be reported separately. If the team develops a fine-tuned adapter, it shall be compared with the unchanged base model on held-out data; training settings and the resulting artifact shall be retained.

Changes in the five prompt dimensions shall be reported as heuristic measures and accompanied by independent review of the plans. Media assessment shall consider image-text alignment, narrative coherence, shot variety and cross-shot continuity separately: a frame-level score cannot establish the quality of a complete video narrative. Human reviewers shall use a declared rubric with randomized presentation. Latency and memory measurements shall identify the machine used. Reports shall include failures, retries, known costs, effect estimates and uncertainty suited to the sample size. Whether the workflow improves results remains a question for the evaluation.

6.5 Acceptance Criteria

1. Valid input completes the core prompt-to-media workflow with clear feedback and downloadable output, and its associated High-priority tests pass.
2. An accepted plan demonstrates wide, medium and close-up procedural compositions; an enabled image route also produces a real image.
3. The external-video objective is tested with an actual synthesized clip, with results separated from videos assembled from still images.
4. Standard and Local Demo modes run with actual processing and recorded model identities; unavailable local capabilities remain clearly identified and restricted.
5. The stated checks for timing, style propagation, retry limits, partial results and provenance are satisfied.
6. The 50-prompt study and stated standard/local comparison report actual findings, missing observations and limitations. Any optional fine-tuning claim is supported by corresponding evidence.
7. The final report records the status of supporting features and unresolved evaluation targets. Documentation, diagrams and test records describe the submitted system.

6.6 Requirements-to-Test Traceability

Every requirement below is linked to a primary test, while individual scenarios may cover related requirements as well. After execution, the test record shall include the actual result, execution date, tested version and a reference to the evidence.

Table 28: Requirement-to-test traceability

Requirement	**Subject**	**Primary test**
FR-01	Prompt submission	TC-01
FR-02	Linguistic scoring	TC-02
FR-03	Actionable feedback	TC-02
FR-04	Optional dictation	TC-03
FR-05	Bounded clarification	TC-04
FR-06	Creative brief	TC-04
FR-07	Style propagation	TC-05
FR-08	Output dimensions	TC-05
FR-09	Workflow choice	TC-06
FR-10	Media route choice	TC-06
FR-11	Narrative decomposition	TC-07
FR-12	Retrieval-grounded planning	TC-08
FR-13	Intent preservation	TC-09
FR-14	Scene continuity	TC-10
FR-15	Storyboard review	TC-07
FR-16	Procedural rendering	TC-11
FR-17	Image synthesis	TC-12
FR-18	Video synthesis	TC-13
FR-19	Background execution	TC-14
FR-20	Duplicate protection	TC-14
FR-21	Progress and recovery	TC-15
FR-22	Controlled failure handling	TC-16
FR-23	Visual feedback	TC-17
FR-24	Narration	TC-18
FR-25	Common media timeline	TC-18
FR-26	Post-processing	TC-19
FR-27	Colour treatment	TC-19
FR-28	Preview and download	TC-20
FR-29	Regeneration records	TC-21
FR-30	Saved history	TC-21
FR-31	Prompt versions	TC-21
FR-32	Execution-mode control	TC-22

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Requirement-to-test traceability (continued)

Requirement	**Subject**	**Primary test**
FR-33	Consistent mode routing	TC-22
FR-34	Local model readiness	TC-23
FR-35	Local retrieval	TC-23
FR-36	Local capability profile	TC-24
FR-37	Model provenance	TC-24
FR-38	Adaptation experiment	TC-25
FR-39	Configuration and readiness	TC-23
FR-40	Comparative evaluation	TC-25
NFR-01	Responsiveness	TC-26
NFR-02	Service latency	TC-26
NFR-03	Bounded execution	TC-16
NFR-04	Resource efficiency	TC-16
NFR-05	Reliability	TC-16
NFR-06	Usability	TC-27
NFR-07	Accessibility	TC-27
NFR-08	Security	TC-28
NFR-09	Privacy	TC-24
NFR-10	Maintainability	TC-29
NFR-11	Portability	TC-29
NFR-12	Media compatibility	TC-18
NFR-13	Reproducibility	TC-25
NFR-14	Budget control	TC-16
NFR-15	Scalability	TC-14
NFR-16	Recoverability	TC-30
OR-01	Data integrity	TC-28
OR-02	Version and run attribution	TC-21
OR-03	Retention and cleanup	TC-30
OR-04	Ethical use and provenance	TC-25
OR-05	Documentation	TC-29

A Glossary

Table 29: Definitions, acronyms and abbreviations

Term	Definition
SRS / FYP	Software Requirements Specification / Final Year Project.
NLP	Natural Language Processing: computational methods for analyzing and generating language.
LLM	Large Language Model: a pretrained model for language understanding and generation.
MAS / agent	Multi-Agent System / a specialized software role within a coordinated workflow.
RAG	Retrieval-Augmented Generation: supplying relevant retrieved passages as context for generation.
Creative brief	Structured clarification of intended subject, action, setting and style.
World state	Character, setting and style details shared across a storyboard.
Storyboard / shot	An ordered visual plan / one scene unit specifying description, camera and duration.
Embedding / FAISS	A model-specific numerical representation, with sentence embeddings supporting semantic comparison [4] / the vector-index library used for retrieval.
API / adapter	Application Programming Interface / a component mapping a shared task contract to a service or runtime.
Standard mode	The default profile combining hosted inference with the selected compatible media route.
Local Demo	A prepared profile that runs supported components on the local device.
Fine-tuning	Documented training that updates model parameters or an adapter; prompting, RAG and quantization alone do not qualify.
Quantization	Use of reduced numerical precision for model values to alter resource requirements.
TTS / beat	Text-to-Speech / a timed unit of narration or silent action within a shot.
FPS / VTT	Frames per second / WebVTT timed caption format.
Job / attempt	A scheduled work unit / a separate, identifiable execution of a generation request.
Provenance	Recorded input, settings, component/model identities and conditions for an artifact or result.

B Analysis Models

These models complement the context diagram, data flow diagrams, component architecture, use-case diagram and entity–relationship diagram presented earlier. They describe required behaviour; internal class design belongs to the design document.

B.1 System Sequence Diagram

Figure 8 is the system sequence diagram for the main scenario of UC-01 to UC-08. VidCraft appears as a single black-box lifeline. The diagram therefore shows only the system events exchanged with the Creator and the requests the system makes to the secondary actor, Model Services. Solid arrows are requests and dashed arrows are replies. The *opt* fragment covers clarification (UC-02). The *alt* fragment separates the single-image workflow (UC-05) from the storyboard workflow (UC-04, UC-06). The *loop* fragment polls progress with the retained job handle until the job reaches a terminal or expired state (UC-07). Downloading the results (UC-08) completes the scenario.

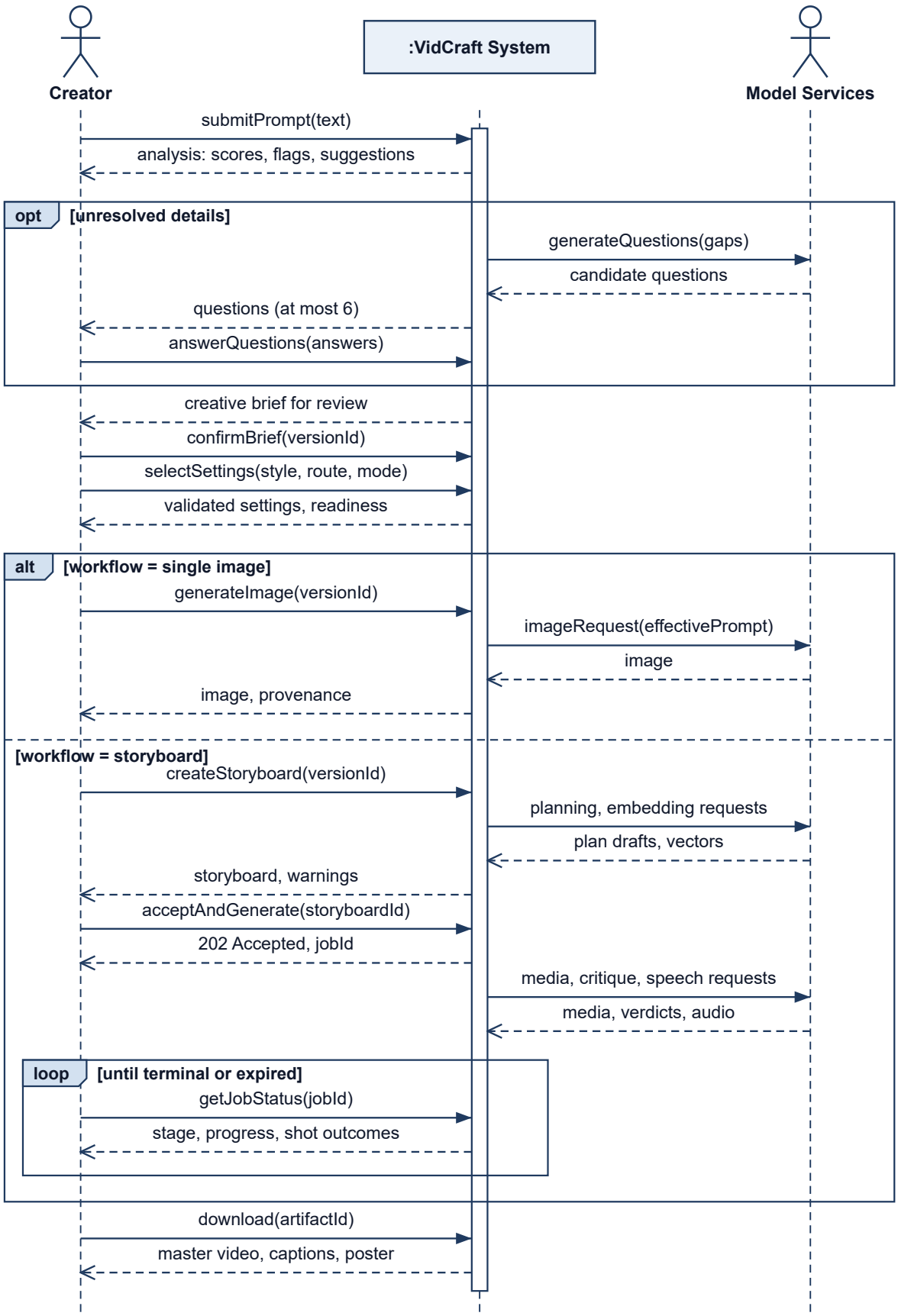


Figure 8: System sequence diagram of VidCraft

B.2 Prompt Refinement and Multi-Agent Planning Sequence

Figure 9 opens the black box for the part of the system that carries the project's main contribution, and it is the only detailed sequence diagram retained. The Workflow Controller analyzes the prompt, asks clarification questions when needed and builds the brief. In the loop, the Screenwriter returns three to five shots with a world state and the controller validates the plan. The Cinematographer then retrieves technique passages (top-$k = 3$) with the matching encoder and refines the shots. Next, the controller embeds the brief and the narrative with the same encoder and checks intent preservation. The loop allows the first draft and at most two revisions, and it ends with a pass, a low-score warning or an UNCHECKED status. Explicit user style remains in the shared state, and every inference call follows the task's execution profile.

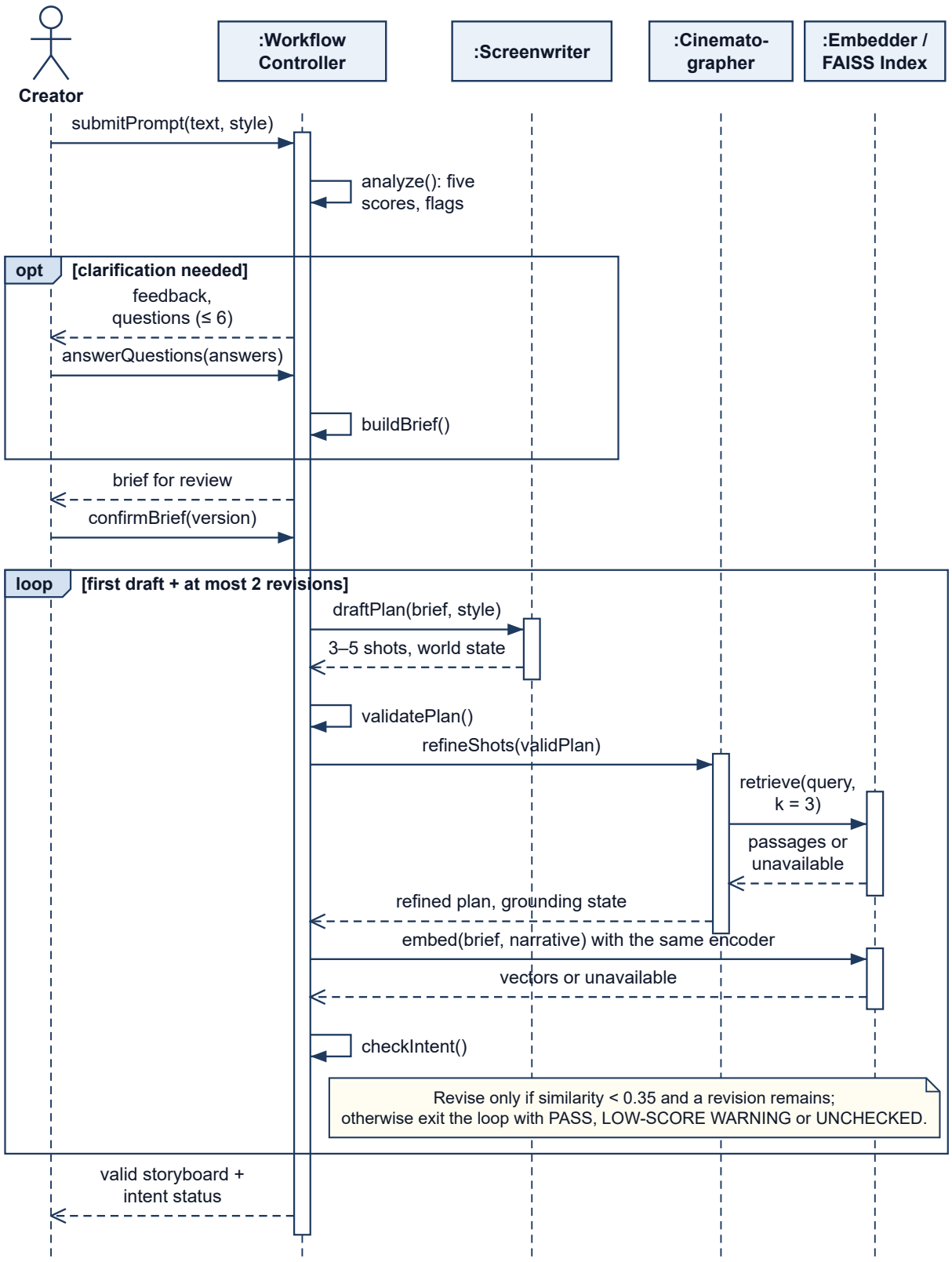


Figure 9: Prompt refinement and multi-agent planning sequence

B.3 Activity Model

Figure 10 traces the complete workflow with one initial node and one activity final node. Decision nodes cover input validity, clarification, capability readiness, the output workflow, plan acceptance, bounded quality retries (at most two regenerations per shot), remaining shots and the presence of usable shots; each outgoing flow carries a guard. Optional narration and post-processing precede the recorded outcome: complete, partial, no output or failed. A request that is not ready, or a storyboard that the Creator rejects, reaches the same final node through a merge node; the Creator then submits a new request after correcting it.

Selecting Local Demo triggers a readiness check of the local runtime, model, index and renderer, and a missing resource follows the *not ready or invalid* branch. The selected execution profile governs every model call in the activity. Hosted inference requires a deliberate switch back to Standard mode.

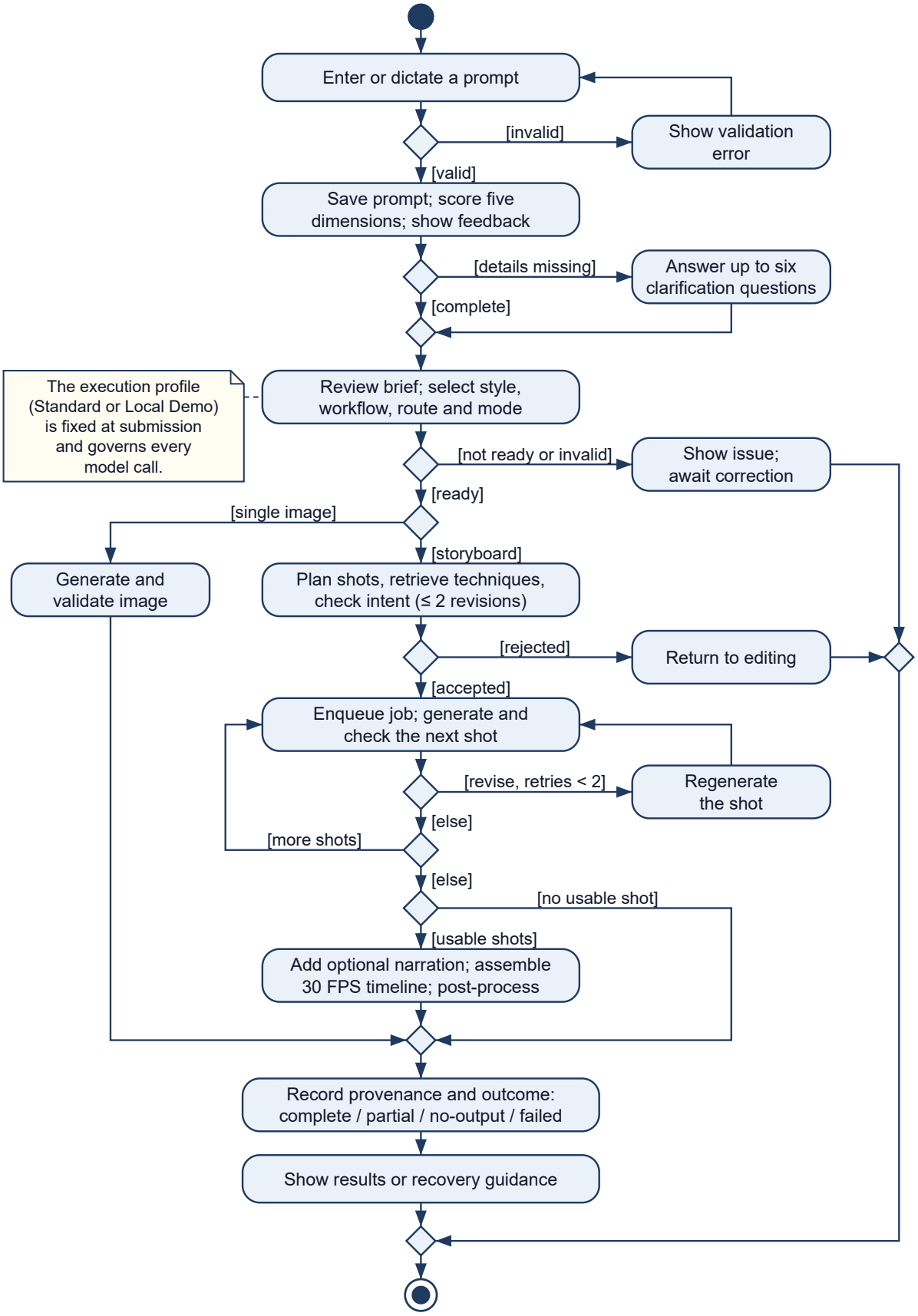
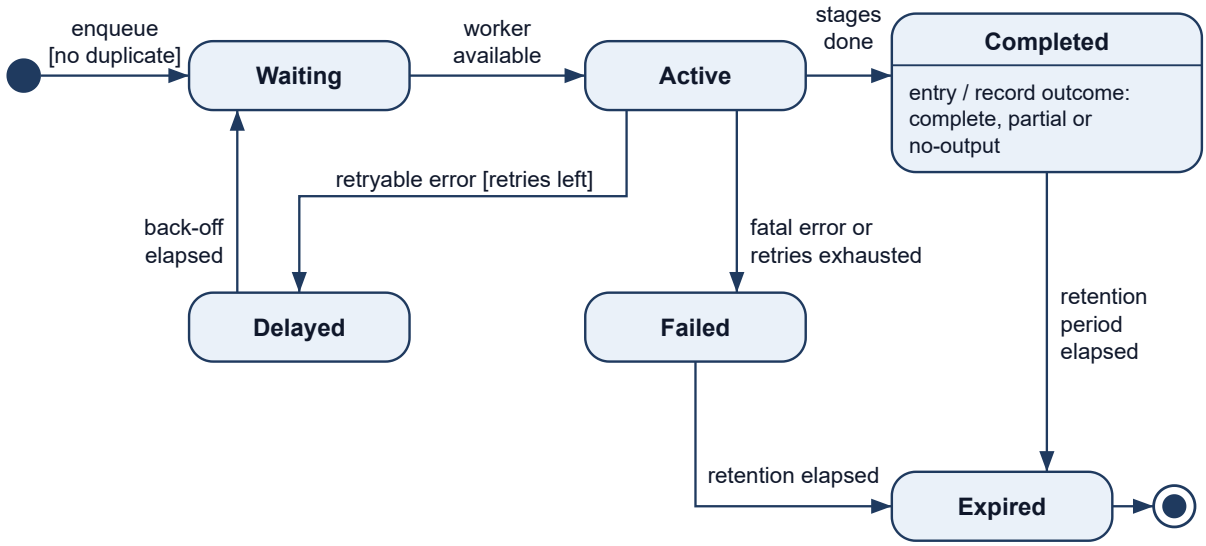


Figure 10: Activity diagram of the VidCraft workflow

B.4 State-Transition Models

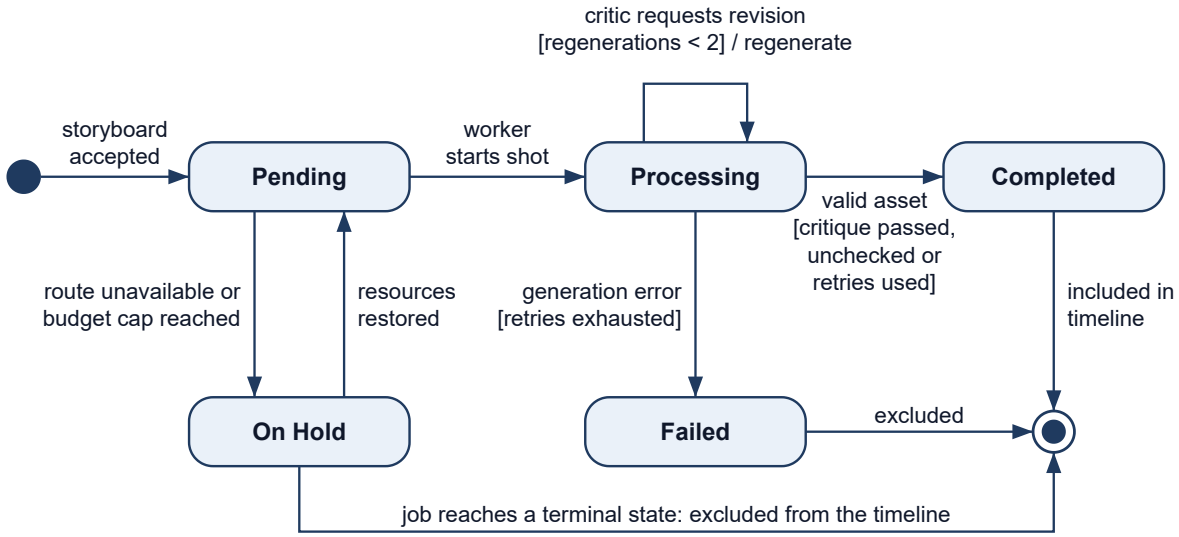
The state-transition diagrams formalize the processing states listed in the business rules. Figure 11 follows one generation job from admission to expiry. A duplicate start request returns the existing job handle instead of creating a new job (FR-20). Retryable errors pass through the Delayed state within the retry cap (NFR-03). A completed job records whether its output is complete, partial or absent (FR-21, FR-22), and Expired marks a record whose retention period has elapsed (OR-03).

Figure 12 follows one storyboard shot. Critique may regenerate a shot at most twice (FR-23). Only completed shots contribute to the common timeline; failed and held shots add no interval (FR-25).



Completion of the queue job does not by itself mean a video exists: the recorded outcome distinguishes complete, partial and no-output results. A duplicate start request returns the existing job handle instead of a new job.

Figure 11: State-transition diagram of a generation job



Only completed shots contribute frames, narration and captions; failed and held shots add no timeline interval (FR-25). Each included shot uses max(1, round(duration × 30)) frames at 30 FPS.

Figure 12: State-transition diagram of a storyboard shot

B.5 Deployment Model

Figure 13 locates the run-time components of the controlled academic deployment. The client computer runs the browser studio. The local or laboratory host runs the Node.js environment (gateway, worker, media adapters and Remotion renderer), the Python AI service, MongoDB, Redis, FFmpeg, the FAISS index, managed media storage and the optional local inference runtime. Hosted model services lie outside the host boundary and are reached over HTTPS. Public/shared hosting is included only where configured with the required access controls; the controlled academic deployment does not require shared or public hosting.

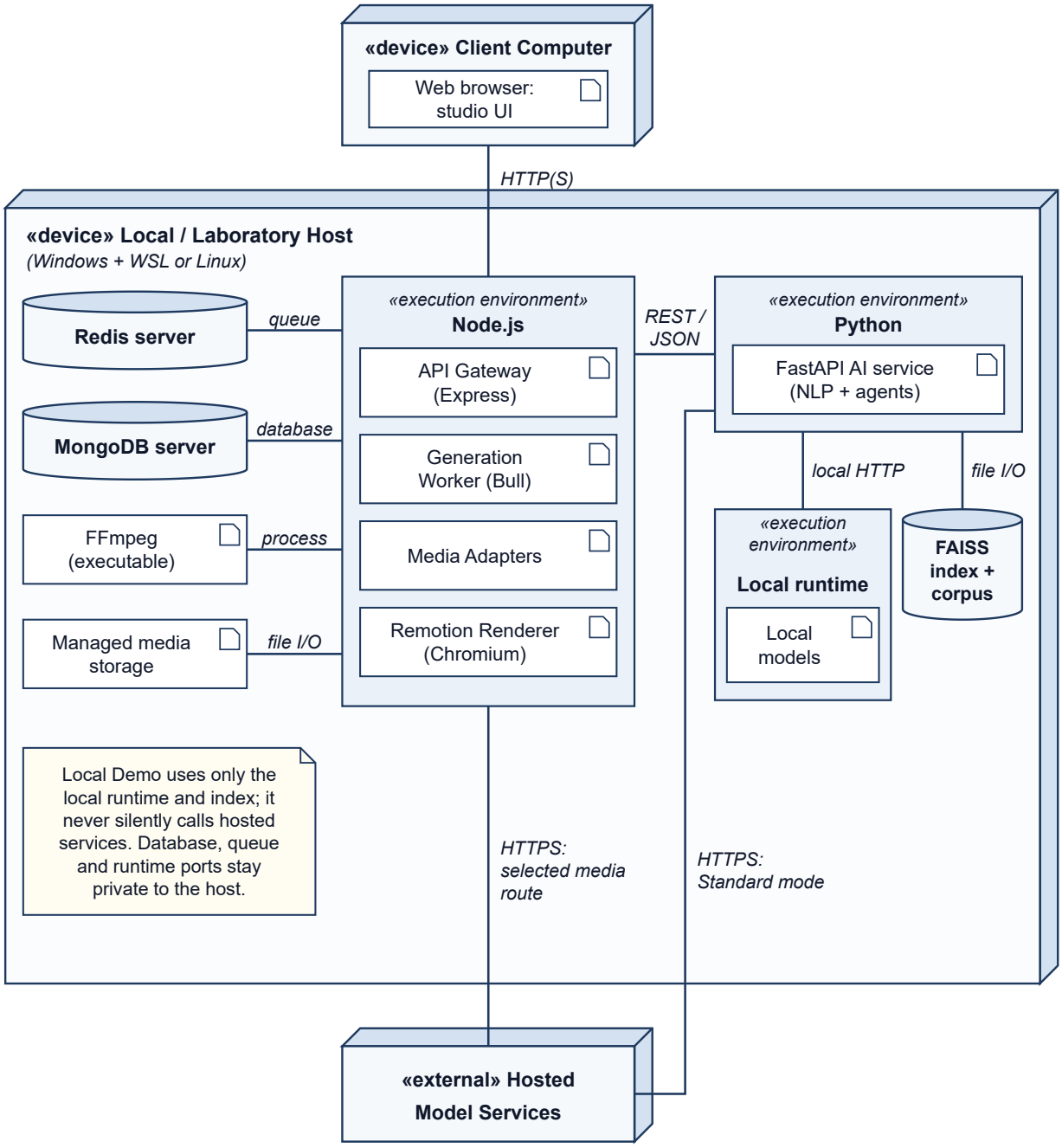


Figure 13: Deployment diagram of VidCraft

C To Be Determined List

These deployment and experimental choices shall be finalized before the relevant acceptance tests. Their resolution shall leave the core workflow unchanged.

Table 30: Outstanding configuration and evaluation decisions

ID	Decision	Resolution criterion
TBD-01	Local model/runtime, quantization and the actual machine's resource limit.	Valid structured output within measured capacity and the configured timeout.
TBD-02	Whether to train a task-specific adapter and which licensed data/task to use.	If pursued, a training artifact, disjoint splits and base-versus-adapted evaluation.
TBD-03	Final video-capable model route, budget and supported dimensions.	An actual clip and a temporal-content assembly test.
TBD-04	Retention period and backup location for ordinary and assessment data.	Documented cleanup/restore procedures that protect assessment records.
TBD-05	Human assessment rubric/raters and the paired standard/local comparison set.	A frozen protocol, declared sample counts and reproducible records.

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